University of Hamburg

Master Thesis

From Sectoral Price Shocks to Overall Inflation: Insights from an Input-Output Framework

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Introduction

After more than a decade of subdued price dynamics and expansive monetary policy, inflation has returned to the center of economic and political debate (Carstens, April 2022). In the Euro Area, inflation rates surged to levels not seen in decades, driven first by the COVID-19 pandemic and later by the geopolitical fallout of the Russian invasion of Ukraine. These developments marked a clear break from the low-inflation environment that had prevailed since the global financial crisis and the time of the*Great Moderation* before. The aftermath of both scenarios challenged the notion of guaranteed price stability and exposed the limits of aggregate monetary policy and demand management to tame price increases.

Recent inflation episodes in the EU and globally have been driven by shocks on the real side of the economy. Asymmetrical, industry-level shocks play a much larger role for overall price levels, according to recent scientific contributions, compared to monetary expansion or other factors (Ferreira et al., 2024; Ipsen et al., 2023; Weber et al., 2024). In particular, the energy crisis revealed how a concentrated supply-side shock can propagate unevenly through production networks, inducing widespread price effects (Cucignatto et al., 2023).

Figure 1 supports the observed, asymmetric nature of price increases on the industry level. Orthodox inflation models focus on aggregate indicators such as demand, supply, wages, and interest rates (Eickmeier & Hofmann, 2025). Given the asymmetric impact on prices across sectors within the economy, such aggregate modeling might be insufficient. Capturing structural aspects, the topology of shocks so to say, can significantly improve the understanding of inflation processes. Thus, it can help to develop and diversify adequate policy for containing inflation. This underscores the need to adopt a more granular and structural analytical perspective, in contrast to models that attribute inflation mainly to shifts in economic aggregates like money supply. In this context, the Leontief input–output (IO) framework by Vassily Leontief (1936) offers a powerful framework for tracing how sectoral price shocks translate into systemic inflationary pressures. Weber et al. (2024) rekindled the interest in this approach, which was subsequently picked up and extended by Ipsen et al. (2023), Konczal and Lusiani (2022), Ferreira et al. (2024) and others.

This thesis contributes to the growing body of literature that links micro-level shocks to macroeconomic outcomes via production networks. Building on the approaches of Weber et al. (2024) and Ipsen et al. (2023), the following research investigates the role of

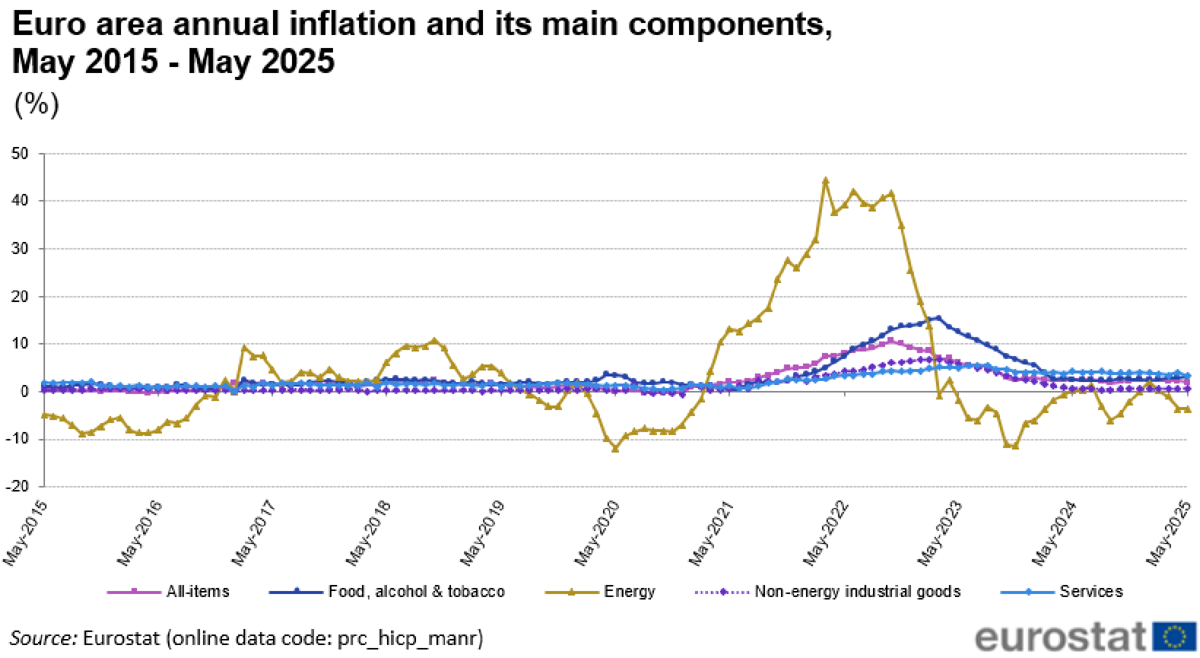


Figure 1: ***Euro Area Inflation and its main components.*** *Energy and food sector price increases were significantly more pronounced. (Source: Eurostat, 2025)*

so called *systemically significant*[1] sectors in shaping inflation dynamics[2]. For the analysis, price shocks for each industry and country available will be induced in the FIGARO[3] IO tables, provided by Eurostat (Eurostat, 2024a, 2024b), expressed in monetary values, which allows to observe the endogenous price effects on other industries. By leveraging consumer demand data from the IO tables, synthetic consumer price indices (CPIs) are constructed to translate relative price changes into estimated impacts on overall inflation. The objective is to identify industries whose price changes disproportionately influence the aggregate price level and to examine how inflation exposure differs across countries and sectors, offering insights into recent inflationary pressures and the adequacy of policy responses. The second part of the analysis will specifically focus on the recent price shock in natural gas, using it as a case study.

Analytically, the focus lies on sectoral, relative price shocks with systemic potential, in other words, examining how such shocks propagate through the economy via IO linkages. This perspective questions the sufficiency of aggregate monetary interpretations and highlights the need for more granular inflation diagnostics. It also challenges the classic definition of the term *inflation*.

[1]A concept originally introduced by Hockett and Omarova (2016).

[2]The full implementation of the analytical framework, including data pipelines, models, and visualizations, is available at https://github.com/DaBurb/Masterthesis-Burbach-Economics.

[3]Full international and global accounts for research in IO Analysis - FIGARO

1.1 The Taxonomy of Inflation: A Misused Term?

The inflationary episodes following the COVID-19 pandemic and the war in Ukraine significantly affected global price dynamics. However, excessive use of the term *inflation* can obscure the nature and origin of price increases. As Humpage (2008) argues, inflation is one of the most misused terms in economics–frequently applied without regard to the underlying causes and often carrying normative implications for policy, particularly the expectation that central banks should raise interest rates to cool aggregate demand.

From a macroeconomic perspective, inflation refers to a process of sustained and accelerating increase in all prices and wages across the economy over time (Hoflmayr, 2022). Tooze (2021) adds more explicitly, that inflation, in the proper macroeconomic sense, excludes idiosyncratic supply shocks. Flassbeck (2021) outlines, that "inflation is not caused by a specific event or a specific shock, but it is always the result of a process in which wages play the decisive role of driver". Hence, the term *inflation* should be used carefully; applying it to isolated or short-lived price changes risks misdiagnosis and misguided policy. This highlights a conceptual problem: recent price increases were highly uneven and rooted in specific disruptions–such as mobility restrictions during the pandemic (Canton et al., 2021) or shocks to natural gas and wheat markets triggered by the war in Ukraine (Aizenman et al., 2024). Bernanke and Blanchard (2024, p. 38) state, that while labor markets were tight, it had little effect. Thus, the recent episodes would not fall under the definition by Tooze or Flassbeck.

Moreover, recent shocks were largely temporary in nature. Many pandemic-related restrictions were lifted, and firms adapted by substituting away from risk-laden inputs–most notably, from Russian pipeline gas toward LNG alternatives (Moll et al., 2023). What unfolded was not a uniform, demand- or wage-driven inflation, but rather a series of *relative and temporary price shocks* (Bernanke & Blanchard, 2024). Importantly, this does not generally preclude the possibility for such relative price shocks to trigger persistent inflation, if, for instance, workers start to demand higher wages following cost pass-through by firms. As stated above, this was not the case. Figure 1 supports, that inflation did not become persistent. The key insight is, that relative price shocks can trigger persistent inflation, given circumstances described in the next chapter, but do not fall under the term *inflation* themselves.

Accordingly, this analysis focuses on relative price shocks, without assessing their persistence over time. For convenience, this analysis uses the term inflation synonymous to overall price increases. But it does not imply *a persistent process of increasing prices and wages* in this work. The following description of related work will underline this notion.

1.2 Related Works

The thesis is inspired and based on a growing body of literature, which analyze the effects of sectoral price shocks on the overall price level. This review will present central contributions.

Weber et al. (2024) argue that inflation surges post-pandemic are structurally shaped by the characteristics of production networks, rather than uniform cost increases or monetary trends. In doing so, they identify systematically vulnerable industries within the production system, aiming to expand the economic policy toolbox beyond aggregate demand management. Using recent US data for the IO model, they simulate sectoral price shocks and quantify their inflationary impact. The method builds on a modified Leontief price model in which individual sectors, as well as a group of commodities, are rendered exogenous and shocked based on their observed price volatility, allowing the simulation of sector-specific disturbances. The resulting changes in endogenous prices are then aggregated into overall consumer inflation using synthetic CPI weights. The price volatility-based shock scenario is used to identify what they term the *latent systemic significance* of sectoral prices, that is, the inflationary risk a sector poses due to both its price volatility and structural position in the production network. This measure offers a general indication of systemic weak spots in the price system. Weber et al. (2024) also incorporate observed US price changes from recent inflation episodes, allowing them to evaluate the actual sectoral contributions during the COVID-19 pandemic and the price changes following Russia's invasion of Ukraine. Their results show that a small group of industries accounts for the greatest structural inflation relevance, broadly grouped into four categories: energy, basic necessities, basic production inputs, and commercial infrastructure (Weber et al., 2024, pp. 310–311). The energy cluster, including petroleum refining, utilities, and oil and gas extraction, stands out due to high price volatility and strong upstream linkages. Basic necessities, such as housing, food and beverages, and agriculture, contribute through large CPI weights or volatile input prices. Among basic production inputs, chemical products emerge as systemically significant due to their dual role in consumption and production, coupled with high ubiquity in supply chains. Finally, wholesale trade forms part of the commercial infrastructure whose systemic importance derives from its centrality in both inter-industry transactions and final consumption, despite exhibiting low price volatility.

Building on this, Ipsen et al. (2023) develop a complementary framework for stress-testing inflation exposure, which constitutes a second cornerstone reference for this thesis. While methodologically similar to Weber et al. (2024), they rely exclusively on price volatility data due to the lack of up-to-date observed sectoral price changes for the European context. The results show comparable sets of sectors regarding systemic significance, compared to the US application by Weber et al. (2024). However, their key innovation lies in the application of differentiated CPI aggregation schemes that reflect structural differences in consumption across macro-regions. By comparing inflation outcomes under EU28-wide CPI weights and an alternative aggregation that distinguishes between core and peripheral EU member states, Ipsen et al. (2023) reveal marked asymmetries in exposure to systemic price shocks. Their core–periphery analysis shows that structurally central economies tend to act as amplifiers of inflationary shocks, passing them to the periphery, where peripheral countries are more vulnerable to externally transmitted cost pressures. While being a valuable source for the application of the model on Europe, this study importantly informs this thesis's investigation into the regional aspect of inflation transmission.

Cucignatto et al. (2023) study the transmission of a gas price shock through the European production network, modeling a 500% increase in gas import prices and tracing its economy-wide impact with a Multi-Regional IO model (MRIO). They consult the OECD Inter-Country Input–Output tables (ICIO). The analysis, focused on France, Italy, and Spain, simulates three policy scenarios: one without price controls, one with uncoordinated national interventions, and one with coordinated price regulation targeting upstream sectors. Their findings reveal that the inflationary impulse of rising gas costs is not solely driven by direct input cost increases, but significantly amplified by firms' efforts to maintain or expand profit margins. This mark-up behavior results in a transmission of shocks well beyond energy-intensive sectors, supporting the interpretation of current inflation as at least partially profit-led. Moreover, the authors demonstrate that coordinated price controls, if strategically applied to systemically significant upstream sectors, can meaningfully dampen inflation while preserving the profitability of most industries. By contrast, fragmented or absent controls tend to shift the burden of adjustment onto consumers, eroding real wages and disproportionately affecting low-income households. The study thus frames inflation as a conflictual distributional outcome, shaped not only by cost pressures but by the institutional and political response to those pressures. This thesis uses the approach for the gas price shock analysis, specifically the first scenario without price controls.

Ferreira et al. (2024) focus on what they call *shockflation* resulting from the recent shocks. Their study is related to to this thesis, since they use FIGARO IO data, similar to this thesis. They are analyzing the inflationary effects of sectoral cost-push shocks across structurally diverse EU regions. Also building on the IO framework introduced by Weber et al. (2024), the authors apply a modified Leontief model to FIGARO IO tables to identify systemically significant sectors and explore inflation asymmetries between core, southern periphery, eastern periphery, and financial hub countries. Their results differ substantially from those of Ipsen et al. (2023), which could be based on their use of sectoral volatility data from the EUKLEMS database[4]. Ferreira et al. (2024) highlight energy, wholesale

[4]EU level analysis of capital (K), labour (L), energy (E), materials (M) and service (S) inputs

trade, housing, and financial services as key inflation drivers. Compared to other studies, their estimates are more pronounced and yield a different sectoral structure, though the identified sectors are similar. Regarding the interpretation of results, they align with Cucignatto et al. (2023), emphasizing a distributional conflict perspective in which firms preserve or expand profit margins amid shocks, contributing to inflation beyond what input costs alone would suggest.

Beyond the foundational studies discussed above, a growing literature highlights how inflationary dynamics are shaped by sector-specific shocks, network linkages, pricing behavior, and structural asymmetries. Several recent contributions explore how upstream shocks, particularly in energy and key inputs, propagate through production networks and lead to persistent inflation. Afrouzi et al. (2024) demonstrate that relative price changes in upstream sectors can act as aggregate supply shocks, especially under nominal rigidities and strong IO linkages. Bilgin and Yilmaz (2018) analyze inflation connectedness, finding that system-wide inflation in US is driven by upstream manufacturing industries such as metals and chemicals, using a Diebold-Yilmaz network framework. Lafrogne-Joussier et al. (2023) confirm that positive energy cost shocks fully pass through to producer prices, while negative ones lead to only limited easing, underscoring asymmetric cost transmission. Again, upstream sectors exert significant influence in the pricing system. This also supports the notion of profit inflation, since firms appear to increase their profit margin on the long run, if they pass increases but not decreases in prices. Baqaee and Farhi (2019) provide a theoretical foundation for the nonlinear propagation of sectoral shocks through production networks. Extending Hulten's theorem, they show that microeconomic structures such as substitution elasticities and network position critically shape macroeconomic outcomes. In particular, their results demonstrate that negative productivity shocks in key sectors—such as energy—can generate amplified aggregate effects due to network nonlinearities and input complementarities.

Thus, inflation persistence cannot be understood without accounting for the behavior of firms and the expectations of households. Weber and Wasner (2023) introduce the concept of *sellers' inflation*, arguing that firms with market power used supply shocks as an opportunity to increase markups. They identify several characteristics of firms in their pricing behavior, which lead to a semi-concerted reaction in the case of price shocks, and increased profit margins. Acharya et al. (2023) provide supporting evidence that supply-chain disruptions raised household inflation expectations, which allowed firms to maintain elevated markups even after bottlenecks eased. This behavioral feedback loop contributes to broad-based inflation and amplifies structural characteristics like sectoral concentration and pricing power. Stiglitz and Regmi (2023) echo this interpretation, framing current inflation as a supply-side phenomenon that cannot be addressed by monetary tightening alone, but calls for institutional intervention.

Other studies emphasize how global and local inflationary forces interact. Bajraj et al.

(2023) document that while services inflation is largely shaped by global dynamics, goods inflation tends to be driven by local factors. Hansen et al. (2023) attribute the euro area's post-pandemic inflation primarily to rising import prices and corporate profits rather than wages, highlighting the external nature of recent cost shocks and the domestic pass-through via firm behavior.

Several studies also expand the methodological toolkit for inflation analysis. Przybyliński and Gorzałczyński (2022) use an IO decomposition to quantify the relative contributions of imports, value added, and intermediate consumption to CPI trends. Acemoglu et al. (2012, 2016) provide the theoretical backbone for many of these approaches, showing how aggregate volatility can arise from sectoral disturbances when the economic network is sufficiently interconnected.

1.3 Research Questions and Structure of the Thesis

This thesis builds on the premise, that not all prices matter equally for inflation. The need for differentiation on the term inflation was outlined before. Having further established the relevance of industry level price shocks for overall inflation by reviewing foundational literature, this leads to several questions, which will structure this thesis.

First, since not all prices matter the same, literature suggests that certain prices may play an outsized role in triggering or amplifying inflationary dynamics due to their volatility and position in the production network. This leads to the first question:

> **(1)** *Are there systemically significant sectors in the EU28 whose prices disproportionately affect inflation during an economic shock?*

Second, beyond identifying such sectors, this thesis analyzes the structural composition of inflation across sectors for the EU28 as a whole. It investigates which sectors consistently drive inflation when averaged across countries and how their inflationary effects are distributed. Are these effects widespread across many countries, or driven by only a few dominant country-sector combinations? This leads to the second question:

> **(2)** *Which sectors are structurally central to inflationary dynamics across the EU28, and how concentrated or diffuse are their contributions across member states?*

Third, this thesis investigates the geographic dimension of systemic price shocks by identifying clusters of structurally similar countries within the EU28 using network-based community detection. It then evaluates how inflationary shocks originating in one regional cluster affect others via production linkages and differentiated CPI baskets. This leads to the third research question:

> **(3)** *How do inflationary shocks propagate between regional clusters within the EU28, and which regions act as amplifiers or absorbers of systemic price shocks?*

Finally, this thesis examines how the inflationary relevance of sectors evolves over time. By applying the same systemic shock scenario across multiple years, it traces temporal patterns in sectoral inflation impact and investigates whether structural inflation vulnerabilities are persistent or shifting. This leads to the fourth question:

> **(4)** *How does the systemic inflation relevance of sectors in the EU28 evolve over time, and do persistent inflation drivers emerge?*

In addition to the systemic analysis based on price volatility, this thesis simulates the transmission of a stylized, but reality oriented, gas price shock through the EU28 production system. The aim is to assess how an extreme cost-push disturbance propagates through sectoral and regional structures, and characterize the transmission of the 2021–2022 gas shock. This motivates a fifth research question:

> **(5)** *How does a gas price shock propagate through the EU28 production network, and which upstream sectors and countries exhibit the highest inflationary exposure to such a scenario?*

To address these questions, the thesis proceeds in two main analytical parts. The first part focuses on identifying systemically significant sectors using a price-volatility-based shock scenario derived from a modified Leontief price model. Drawing on the approach by Weber et al. (2024) and its adaption for EU data and extension for regional analysis by Ipsen et al. (2023), it estimates the inflationary risk posed by sectoral prices in terms of their *latent systemic significance*.

The second part is a stylized case study of a specific episode: the 2021–2022 surge in gas prices. Building on the framework by Cucignatto et al. (2023), it models a gas price shock propagating through the EU28 production network using inter-country IO data. It then evaluates the resulting inflationary impact across countries and consumption categories, considering regional CPI weighting schemes.

Together, both parts provide a coherent answer to the research questions: they show that inflationary dynamics are shaped by the structural configuration of the economy and that relative price shocks can trigger system-wide effects with unequal regional burdens.

Chapter 2

Theoretical Foundations and Perspectives on Inflation

Inflation, while a macroeconomic phenomenon, originates in firm- and sector-level price decisions in response to changing costs or expectations. This raises a core theoretical challenge: how do micro-level price shocks aggregate into economy-wide inflation?

The issue lies at the intersection of microeconomics, focused on decentralized agent behavior, and macroeconomics, which analyzes aggregate outcomes like inflation and employment. Keynes formalized this divide between micro and macro level, which still shapes inflation theory and policy today (Snowdon & Vane, 2005, pp. 21–24).

This thesis aims to contribute to the disaggregated Post-Keynesian/Structuralist perspective, which focuses on how sector-specific price changes—particularly in systemically significant sectors—can shape macroeconomic inflation dynamics.

2.1 Classical and Neoclassical Theories of Inflation

Among the diverse theoretical frameworks that attempt to explain the origin and propagation of inflation, classical and neoclassical approaches provide some of the earliest (classical) and most popular (neoclassical) models. For the prior, inflation roots in real-sector dynamics, while the latter attributes it to changes in the money supply. While differing in emphasis, both frameworks share the assumption of full-employment and self-regulating markets on the long term, and that inflation emerges from system-wide mechanisms, rather than from heterogeneous sectoral processes. As such, they represent conceptual starting points for understanding inflation's macro-level logic under specific circumstances –yet, as this chapter will argue, their limitations in accounting for structural asymmetries or sector-specific dynamics justify the need for alternative perspectives.

2.1.1 Classical Economics

While classical theory is not monolithic, it generally separates the theory of value from the theory of output (Green, 1982, pp. 61–62). The quantity of money does not affect real economic variables (output, employment, price ratios) in the long term. Consequently, money is considered neutral in the long run (Samuelson, 1968, p. 2; Snowdon and Vane, 2005, pp. 38, 50–54). In the classical framework, *natural price* refers to the price of a commodity that reflects its production cost under "normal" conditions. Here, production costs determine commodity values, with prices gravitating toward their natural price under competitive conditions. The *market prices* of commodities gravitate toward this price, while fluctuating due to temporary factors. The gaps between market prices and natural prices then describe either excessive or deficient effectual demand. Inflation arises in such situations when excessive demand is prevalent.

Output, on the other hand, is assumed to be dependent on the level of capital accumulation, and is treated as a separate constant, or variable. Output theory relies on Say's Law, which assumes production creates its own (excess) demand (Sowell, 2017). Under this assumption, unemployment is always temporary or transitory. General overproduction was deemed impossible. Hence, from the classical perspective, monetary factors and deficient demand are not valid explanations for sustained unemployment–output and employment are ultimately determined by real supply-side factors, particularly capital accumulation and population growth.

The implications for inflation are twofold. First, in classical inflation theory, relative price and, ultimately, the overall price level are determined by the *real sector* (Green, 1982, p. 64). In the larger narrative of this theoretical background section, the real sector can be understood as the micro level. Due to the focus on the theory of value, firms (and sectors) determine prices based on the costs of production. Thus, prices and profit rates are independent of monetary supply. Money supply facilitates (if correctly determined), but does not influence long-run prices. But, secondly, the term *long-run* indicates a relevant differentiation, which alters the statement earlier: The relative price level and the overall price level are determined by the real sector *on the long-run.* How does this affect inflation over time? To show this, the essential equation in classical and neoclassical is the equation of exchange, also known as quantity equation:

$$MV = PY \tag{2.1}$$

M is money supply, V represents the *velocity of money*, which captures the average number of transactions of a unit of money in a given period, P represents the price level, and Y represents total production[5]. From a classical perspective, due to Say's identity, V and Y are assumed constant. Since the real sector is determining the price level long-term, thus, P is the independent variable, and M is the dependent variable (Green, 1982, p. 63). Thus:

$$\text{If } \overline{V} \text{ and } \overline{Y}, \text{ then on the long-run } \uparrow P \;\Rightarrow \uparrow M \tag{2.2}$$

David Riccardo permeated the acceptance of Say's law also in the short term, which was debated previously (Green, 1982, pp. 74–76). Consequently, monetary expansion

[5]Sometimes also Q or T for volume of commodity transactions.

(short-term) could only result in inflation, since velocity and production are assumed constant. Thus:

$$\text{If } \overline{V} \text{ and } \overline{Y}, \text{ then on the short-run } \uparrow M \ \Rightarrow \uparrow P \tag{2.3}$$

Overall, inflation in the classical theory is primarily driven by changes in the real sector, i. e. the micro level, rather than monetary changes, i. e. the macro level. The real sector determines the monetary requirements. Thus, inflation is a result of changes in the real sector. But in the short-term, monetary expansion can revert the direction the quantity equation works, and generate price increases. The focus of classical economists on factors rooted in the firm level might also be partly explained by the lack of macroeconomy as a dedicated discipline in classical economy, this divide was later introduced by Keynes. Regarding the role of the state, the neutrality of money condition does render interventions by the government neither necessary, nor desirable (Snowdon & Vane, 2005, p. 37).

2.1.2 Neoclassical Inflation Theory

Neoclassical economists, e. g. Monetarist school and later the New Classical economists, built upon the classical foundation, while providing a more focused explanation of short-run fluctuations and the dynamics of inflation. Like their classic predecessors, they upheld the view that money is neutral in the long run, but then committed to the notion, that inflation is ultimately a monetary phenomenon. Furthermore, they sought to explain why monetary changes *might* influence real output in the short run and how expectations play a crucial role in that process.

A pivotal development came in the 1960s with the work of Friedman (1968), who introduced the concept of a *natural rate of unemployment* (Gordon, 2018). The natural rate of unemployment is the level of unemployment at which the labor market is in equilibrium, also referred to as the NAIRU (Non-Accelerating Inflation Rate of Unemployment), to which the economy gravitates in the long-term. This concept does have similarities with the concept of the natural price by classical economists, of course applied on (un)employment, not prices. However, Friedman and Phelps accepted that there could be a short-run inverse relationship between inflation and unemployment (depicted by the Phillips curve), but they argued that any attempt to exploit this trade-off systematically would be temporary at best (Snowdon & Vane, 2005, p. 178).

Their expectations-augmented Phillips curve showed that unanticipated inflation temporarily reduces unemployment, but expectations adjust, making the long-run Phillips curve vertical at the natural rate. Consequently, there is no permanent trade-off between inflation and unemployment: if authorities try to push joblessness below the natural rate, the result will merely be accelerating inflation with no lasting gains in output or employment file. Friedman's analysis reconciled classical long-run money neutrality (this assumption was undermined by Keynesian economics) with observed short-run non-neutrality, by attributing the latter to slow adjustments in expectations (Snowdon & Vane, 2005, p. 178). Money can affect real variables briefly when a change is not fully anticipated, but not in the long run once people catch up to the new inflation rate.

In a sense, monetarists reverted how the quantity equation is read. For classical economists, the prices are based on on the theory of value, which would then determine the optimal money supply, thus, M is rather passive, and dependent on P in the classical theories. The monetarists started to draw pronounced causality of the money supply by central banks, which would then influence price level, thus, M determines the price level P.

The next evolutionary step in neoclassical thought, the New Classical school of the 1970s pushed these ideas to their logical extreme by incorporating rational expectations and rigorous microeconomic foundations into macroeconomic models (Snowdon & Vane, 2005, pp. 219–272). New Classical economists introduced rational expectations, arguing that only unanticipated monetary changes affect real output, as agents adjust wages and prices based on all available information. They do this immediately, for instance following on a statement about monetary expansion by a central bank.

This led to the policy ineffectiveness proposition: anticipated monetary policy cannot affect real variables, only inflation (Snowdon & Vane, 2005, pp. 242–247). The New Classical school emphasized credible, rule-based policy over activist fine-tuning. The prescription was that central banks should focus on price stability (for instance, by committing to a low steady growth rate of the money supply) and not attempt to target employment or output, since any such attempts would either fail or only create volatility if they relied on surprising the public.

The incorporation of individual expectations does imply a bond between microeconomic foundation and macroeconomic outcome. The point is, that individual expectations are assumed to be rational, which reinforces market clearing. In the case of a sectoral price deviation, consumers, since being rational, substitute away to other products. The market goes back to equilibrium, given the money supply is determined correctly. In that sense, the micro aspect does not have significant impact on aggregate inflation. Money remains neutral, and money supply the driver of inflation.

While both classical and neoclassical theories treat money as neutral in the long run, they diverge in how they conceptualize inflation and its causes. Classical theory anchors inflation in the real sector, where competition and production costs determine prices, and the money supply passively adjusts to the needs of output. In contrast, neoclassical and monetarist successors reverse this causality: inflation is seen as a fundamentally monetary phenomenon, driven by excess money growth. Despite integrating individual expectations and optimization into macroeconomic models, neoclassical models retain the belief in frictionless markets and self-correction. They thus continue to treat inflation as a nominal imbalance, best managed through credible monetary rules. This perspective would deem the approach pursued in this thesis unnecessary. While a sectoral shock *might* lead to local short term price deviations, it would not have significant or persistent influence on overall inflation. Even worse, fiscal support or price controls to contain sectoral price fluctuations would harm the process of going back to equilibrium state. In contrast, the justification for the empirical strategy chosen in this work lies in contemporary Keynesian traditions, which integrate possible channels for industry-level shocks to overall inflation.

2.2 Keynesian and New Keynesian Theories of Inflation

Keynesian economics emerged as a direct response to the evident shortcomings of classical theory during the Great Depression. Keynes challenged the assumption that market-clearing mechanisms would automatically ensure full employment and macroeconomic stability, without any government intervention. He argued that focusing exclusively on the long-run equilibrium state is insufficient for describing real-world economic dynamics and leads to inadequate policy prescriptions. This holds particularly true in times of crisis. As Keynes famously remarked, “in the long run we are all dead” (Snowdon & Vane, 2005, p. 54). While Keynes introduced the original ”schizophrenia” (Snowdon & Vane, 2005, pp. 21–24) in economic thinking, by clearly distinguishing between micro-economics and the newly created macroeconomics, this tradition has several evolutionary strands. Before turning to contemporary Keynesian theory, it is essential to outline its foundational principles.

2.2.1 Keynesian Economics

In Keynes’s framework, output and employment are driven by effective demand, meaning the level of aggregate expenditure planned by households and firms. Keynes rejected the concept of supply creating its own demand (Say’s Law), and asserted that aggregate demand can be deficient, leading to under-employment equilibrium (Snowdon & Vane, 2005, p. 69). Keynes formalized this idea through the identity $E = C + I$, with C consumption expenditure and I as investment, where macro-level demand depends on income-sensitive consumption and investment shaped by expectations (Snowdon & Vane, 2005, p. 59). Importantly, income and investment are dependent on another by multiplier effects, increases in I lead to increases in C (Snowdon & Vane, 2005, pp. 59–61). Because there is no automatic mechanism to ensure that all income is spent, aggregate demand can fall short, leading to involuntary unemployment. This poses a macroeconomic failure rooted in microeconomic prudence. In other words, demand (or lack thereof) determines supply in the short run, not vice versa. As a result, it is possible for the economy to settle at an equilibrium with idle capacity and high involuntary unemployment, unless there is some intervention to boost total spending (Snowdon & Vane, 2005, pp. 64–67).

Keynes also fundamentally reinterpreted the role of money and interest rates in the economy. Whereas classical theory posited that the interest rate equilibrates saving and investment to ensure full employment, Keynes argued that interest rates are instead determined by the demand for and supply of money, captured in his concept of *liquidity preference* (Snowdon & Vane, 2005, pp. 69–70). In formal terms, money demand is given by:

$$M^d = L(i, Y),$$

where M^d is the demand for money, i the nominal interest rate, and Y income. Agents hold money for transaction and precautionary motives, particularly under uncertainty. When confidence is low, this preference for liquidity can keep interest rates above levels required to stimulate investment, even with expanding money supply. Keynes calls this a *liquidity trap*. In such situation, monetary policy becomes ineffective, and fiscal expansion may be required to restore aggregate demand. This logic also implies an unstable or volatile velocity of money (Snowdon & Vane, 2005, p. 70).

These insights undermine the classical doctrine of long-run money neutrality (Snowdon & Vane, 2005, p. 63). If aggregate demand shortfalls persist and nominal prices and wages adjust only slowly, then monetary, or nominal changes in general, can have lasting real effects, especially in the short and medium run. Price and wage rigidities, rooted in long-term contracts, institutional norms, or resistance to nominal cuts, prevent the economy from rapidly returning to full employment. As a result, full employment becomes an exceptional case, not a natural market outcome (Snowdon & Vane, 2005, pp. 70, 82–83). Within Keynesian frameworks, inflation is therefore not treated as a purely monetary phenomenon. While Keynes himself did not develop a dedicated inflation theory, his rejection of Say's law, the neutrality of money, and his emphasis on nominal rigidities enabled later theories to explain inflation as emerging from aggregate demand pressures or cost shocks.

Notably, Keynesian theory marked a decisive shift in macroeconomic thought by emphasizing aggregate demand failures and instability, diverging from the classical presumption of self-equilibrating markets. Unlike later neoclassical models, Keynes did not derive macro outcomes from microeconomic optimization, but rather treated aggregate behavior as potentially destabilizing. Individually rational actions, such as firms cutting output in response to weak demand or workers accepting lower wages, do not necessarily lead to collectively optimal outcomes. Keynes pointed out the potential of coordination failures and feedback loops that could trap the economy in persistent unemployment. This shift created what Snowdon and Vane (2005, pp. 21–24) term a "schizophrenia" in economics education: a disconnect between the economic disciplines of microeconomics and macroeconomics.

Monetarist and neoclassical perspectives gained dominance in the wake of the stagflation era, thereby challenging the previously prevailing Keynesian consensus. New Keynesian economists sought to reestablish core Keynesian insights within a framework consistent with rational expectations and microeconomic foundations derived from neoclassical models.

2.2.2 New Keynesian Economics

Thus, the New Keynesian perspective emerged in response to the now dominant monetarist and neoclassical schools, aiming to reconcile Keynesian principles with modern theoretical tools (Goodfriend & King, 1997, p. 246). While New Classical economists emphasized rational expectations and market clearing, they struggled to explain the simultaneous existence of persistent inflation and unemployment (supposed be temporary and self-correcting). New Keynesians retained rational expectations but introduced nominal rigidities. These explain, why macroeconomic fluctuations persist even when agents are rational and forward-looking (Snowdon and Vane, 2005, p. 227; Goodfriend and King, 1997, pp. 249–251).

Two core mechanisms distinguish New Keynesian inflation theory from the neoclassical version: Nominal rigidities and market imperfections. Nominal rigidities arise because firms and workers adjust prices and wages only infrequently, due to menu costs[6], long-term contracts, etc. (Goodfriend and King, 1997, p. 250; Snowdon and Vane, 2005, pp. 366–376). These frictions imply that even small shocks to demand or cost conditions may not trigger immediate nominal adjustments. Instead, firms often postpone price changes, which leads to output and employment fluctuations rather than price-level adjustments. Moreover, firms in monopolistic competition hesitate to lower prices unless others do, amplifying inflation persistence (Snowdon & Vane, 2005, pp. 364–365). Such market imperfections reinforce these dynamics: Price-setting replaces price-taking, which allows for mark-ups that vary with economic conditions or strategic decisions. On the other hand, asymmetric information harms efficient allocation of goods, for instance, consumers might react delayed on market signals. As a result, real shocks or policy changes propagate more slowly through the economy, and inflation emerges as a dynamic process shaped by expectations and slow nominal adjustments at the micro level. These mechanisms provide a microeconomic rationale for persistent inflation, even when agents are rational and forward-looking. In this way, New Keynesians recover Keynes's insight that nominal rigidities create real effects. But they also incorporate microeconomic optimization, due to retaining rational expectations from the neoclassical theory.

[6]The cost of changing prices. The name is derived from the analogy of having to print new menus in a restaurant, when changing prices. Printing new menus generates costs.

Hence, inflation in New Keynesian theory is driven by the interplay of expectations and real activity, formalized in the New Keynesian Phillips Curve (Brissimis & Magginas, 2008, p. 5):

$$\pi_t = \beta E_t[\pi_{t+1}] + \kappa x_t,$$

where π_t is current inflation, $E_t[\pi_{t+1}]$ is expected future inflation, x_t is the output gap, β reflects the forward-looking nature of expectations, and κ captures the sensitivity of inflation to the output gap, which depends on the degree of price stickiness. This formulation implies that inflation is not simply a result of past monetary expansion, but also of how firms and workers anticipate future inflation and adjust their behavior accordingly.

Because prices adjust only gradually, monetary policy can influence real variables in the short run by shaping expectations as well as aggregate demand. A credible commitment to low and stable inflation anchors expectations, aligning wage- and price-setting behavior with the central bank's target (Goodfriend & King, 1997, p. 274). Such a policy reduces uncertainty, thereby helping to prevent inflation ex-ante. Inflation targeting frameworks rely on this mechanism: if agents trust the policy regime, anticipated inflation remains stable, which mitigates both inflation volatility and the employment losses typically associated with disinflationary adjustments.

This reasoning culminated in what is known as the *New Neoclassical Synthesis*, which combines rational expectations and intertemporal optimization (from New Classical theory) with nominal rigidities and imperfect competition (from New Keynesian theory) (Snowdon and Vane, 2005, p. 411; Goodfriend and King, 1997, pp. 255–260). It offers a micro-founded account of inflation in which monetary policy plays a stabilizing role, effective in the short run due to frictions, but ultimately neutral in the long run. Here, inflation arises from the delayed response of prices and wages, and not solely from money supply growth, distinguishing it clearly from the monetarist or pure neoclassical views.

New Keynesian economics thus resolves the tension between micro-level optimization and macroeconomic instability by demonstrating how individually rational price-setting decisions can collectively lead to persistent inflation or unemployment. The theory replaces the neoclassical assumption of frictionless adjustment with a structured account of why markets may fail to self-correct, and why stabilization policy is necessary, even when expectations are rational. A price shock in such a

2.3 Post-Keynesian Perspectives on Inflation

Post-Keynesian economics offers a different explanation of inflation compared to classical and monetarist theories, as well as New Keynesian theories, to some extent. Instead of viewing inflation as primarily a monetary phenomenon or a result of aggregate demand exceeding supply, Post-Keynesians emphasize real-economy factors, especially conflict over income distribution and structural characteristics of the economy, as key drivers. This perspective shifts the focus from aggregative indicators, like money supply or aggregate demand, or abstract equilibrium conditions to the tangible conflicts and cost pressures within the production system. Consequently, micro foundations are crucial for the inflation process in this theoretical perspective. This does not generally preclude macro factors as relevant to inflation, rather, it broadens the view on possible channels and triggers.

Friedman's (1970) well-known statement, that *inflation is always and everywhere a monetary phenomenon* is explicitly rejected; instead, Post-Keynesians, like Eckhard Hein (203-204 Hein, 2024, year) argue that *inflation is always and everywhere a conflict phenomenon*. This statement introduces a dominant theoretical concept among the pluralism of inflation theories of this intellectual superstructure: The conflict theory of inflation.

Inflation in this framework, often called *conflict inflation* for simplicity, is the outcome of distributional battles between different agents, called claimants, in the economy, which try to push through their respective claims on income. The four categories of claimants are: (i) Firms and owners of different forms of capital or land, also generally referred to as capitalists, who aim to maintain or increase their profit shares or rents; (ii) workers, or labor, who seek to protect or raise their real wages; (iii) the government, which exerts its claims through taxation, subsidies, or public sector wages; and (iv) the external sector, where terms of trade, import prices, and exchange rates affect domestic cost structures (Hein, 2024; Rowthorn, 1977). The differentiation of claimants is not always fully identical across all models and applications, but the most important claimants in the private sector are virtually always covered: Firms (or capitalists) and workers (or labor) (Hein & Häusler, 2024; Lorenzoni & Werning, 2023; Rowthorn, 1977; Weber & Wasner, 2023; Wildauer et al., 2023). In the conflict theory of inflation, a persistent inflationary process can only be triggered, if the claims are inconsistent.

Inflation arises when distributive aspirations among claimant groups become mutually incompatible. Each group–firms, workers, the state, and the external sector–forms expectations about its aspired share of income, based on institutional norms, past outcomes, or strategic objectives (Rowthorn, 1977). These *aspiration levels* are not automatically adjusted in response to macroeconomic constraints or to the changing claims of others. When the sum of these aspirations exceeds what is feasible given real output, an *aspiration gap* emerges. Rather than a neutral reallocation, this gap sets in motion a dynamic process of compensatory action: firms raise prices, workers demand higher wages, and the state adjusts taxes or transfers. If no group revises its aspirations downward, the result is a sustained inflationary process rooted in unresolved distributional tensions. Such process is known as a *wage-price-spiral*, or rather *wage-profit-spiral*.

Consider a situation in which no group can expand its claims through growing real output. One group, say, firms or capitalists, manages to increase its distributive share by raising profit margins. This shift manifests as a change in relative prices or wage levels. Two scenarios may then follow. If the other claimants, workers, accept the new distributional configuration, the process leads to a one-off adjustment in the price level-either overall or relative-without triggering ongoing inflation. However, if these groups resist the change and attempt to restore their former share, for example by negotiating higher nominal wages to preserve purchasing power, the aspiration gap opens. What unfolds is a dynamic adjustment process, where each side reacts to the other's move in an effort to close its own gap. As long as these conflicting aspirations persist and are backed by sufficient institutional or market power to push forward own claims, inflation becomes the mechanism through which the struggle over income shares plays out (Hein, 2024, p. 2). Notably, any factor that causes one party to increase its claim on income, say, an aggressive profit target by firms, a surge in raw material costs that firms pass on, a strong union push for higher wages, a hike in consumption taxes, or a jump in import prices, can theoretically be the *trigger* for conflict inflation if other parties refuse to accommodate it.

Now, while this is technically correct, the examples work under the assumption, that the power to defend respective claims is evenly distributed between claimants. In conflict inflation models, the balance of power in wage- and price-setting is crucial. The degree of labor organization and bargaining power versus firms' market power will determine how large the aspiration gap can become and thus how fast prices escalate. For instance, a highly unionized labor force in times of low unemployment may secure real wage increases, where firms in well functioning and competitive markets lack possibilities to increase prices. Conversely, if firms have strong monopoly power, they may hike prices to boost profits, forcing workers to accept real wage cuts.

An important implication of the conflict perspective is that inflation can occur even in the absence of a maximum capacity-utilization, full-employment economy (Hein, 2024, p. 203; Rowthorn, 1977; Lavoie, 2022, pp. 592–593; Lavoie, 2006, pp. 127–130). Classical and neoclassical models assume, that due to flexible substitution of capital and labor, firms will simply decrease or increase quantitative output to meet volatility in demand in the short-run. From this perspective, inflation arises, when firms are not able to meet aggregative demand anymore by increases in output–an output gap emerges. At this point, they will start to increase prices. This is not as relevant for post-Keynesian interpretations. Here, the relevant aspect is the distribution of what is produced in real terms. For example, an external price shock, as they are analyzed in this thesis, can induce price pressure into price system, independent of the rate of unemployment or the capacity utilization. If firms start to pass through prices, workers will likely try to offset the real losses by demanding higher wages. Again, if no agreement is reached, the consistent aspiration gap can induce a wage-profit spiral.

Thus, post-Keynesian inflation theory generally stands in contrast to the mainstream notion of a *natural rate of unemployment* (NRU) or *Non-Accelerating Rate of Inflation*

(NAIRU) (Rowthorn, 1977, p. 222; Lavoie, 2022, p. 593; Lavoie, 2006, pp. 127–130; Stockhammer, 2003). Post-Keynesian theorists argue, that the supposed trade-off between unemployment and inflation is neither stable nor exogenous, and that the Phillips-curve based NRU/NAIRU concepts are too mechanistic. In this view, what is interpreted as a "natural" or "non-accelerating" unemployment rate is simply the outcome of a historically contingent power balance. Consequently, this rate becomes path-dependent: It is endogenous to macroeconomic policy, wage-setting norms, and the broader socio-political environment (Hein & Häusler, 2024; Stockhammer, 2003). Moreover, so called hysteresis effects imply that prolonged periods of high unemployment can themselves alter the structural parameters of the labor market (Logeay and Tober, 2006; Rowthorn, 1977, p. 219). In such a scenario, the position of workers is increasingly weakened; instead of going back into equilibrium due to deflationary tendencies, the NAIRU (or NRU) shifts. This significantly undermines the predictive value of a time-invariant NAIRU, since if it does exist, it is volatile. Thus, for post-Keynesian economists, the Phillips Curve is not a structural, policy-invariant relationship but a reflection of shifting institutional and conflictual dynamics. Therefore, attempts to control inflation through policies aimed at steering unemployment toward a presumed natural rate may rest on flawed theoretical premises and risk entrenching socially suboptimal levels of joblessness (Rowthorn, 1977, p. 237; Weber and Wasner, 2023, p. 207; Stiglitz and Regmi, 2023, p. 337).

In contemporary economies, firms are often in the dominant position to raise prices and expand profit margins, especially in the recent post-pandemic inflation episode (Cucignatto et al., 2023; Glover, Mustre-del-Rio, & Nichols, 2023; Glover, Mustre-del-Río, & von Ende-Becker, 2023; Konczal & Lusiani, 2022; Matamoros, 2024; Nikiforos et al., 2024; Weber & Wasner, 2023). Over the past decades, many markets have become more concentrated and global competition has waned, enhancing the pricing power of large firms (Amiti & Heise, 2025; Autor et al., 2020; Bajgar et al., 2023). At the same time, labor's institutional strength (e.g. union density and collective bargaining) has declined in many countries, limiting workers' ability to secure wage gains in line with prices(Autor et al., 2020). This imbalance meant that when economies reopened after COVID-19, and sectoral supply shocks hit (e.g. energy price spikes, supply-chain bottlenecks), firms were well-positioned to pass higher costs onto consumers, and in absence of wage increases raise their profit share. Such imbalances imply further problems for containing inflation.

2.3.1 Cost-push and Profit-Driven Inflation

With regards to the set foundation of inflation as a distributive conflict between claimants, Post-Keynesians place particular emphasis on *cost-push factors and firms' pricing power* as sources of inflation. In contrast to theories that treat the price level as a (market-clearing) outcome of supply and demand, post-Keynesian price setting evolves around firms (Shapiro & Sawyer, 2003). Post-Keynesians argue that firms set prices as mark-ups over costs, and can adjust these mark-ups strategically, reflecting their interests and market power rather than automatic cost-pass-through or market-clearing mechanisms (Shapiro & Sawyer, 2003). In short: "Prices reflect the interests of firms rather than the conditions of their industries or markets." (Shapiro & Sawyer, 2003, p. 364). "Strategic" implies, that the current position in the market and future goals of a firm are determinant.

Thus, any increase in costs, such as a rise in raw material prices or a negotiated rise in money wages, will push up prices as firms pass on the cost increase to protect their profitability. This is the classic *cost-push* route to inflation, which does not require an overheating economy at maximum capacity, but firms with the ability to raise prices (e.g. due to imperfect competition) and the absence of offsetting policy (like price controls or subsidies) to prevent the cost pass-through.

At the same time, firms may also raise their markup *beyond* what is needed to cover rising costs, especially if market conditions allow it. This can lead to *profit-driven inflation*, or *sellers' inflation* (Cucignatto et al., 2023; Glover, Mustre-del-Rio, & Nichols, 2023; Glover, Mustre-del-Río, & von Ende-Becker, 2023; Hayes & Jung, 2022; Konczal & Lusiani, 2022; Matamoros, 2024; Nikiforos et al., 2024; Weber & Wasner, 2023). In this perspective, the impetus for price increases comes from firms expanding their profit margins. Empirical evidence from the post-COVID and gas price shock period has given credence to this mechanism (Cucignatto et al., 2023; Hayes & Jung, 2022; Matamoros, 2024; Nikiforos et al., 2024). The inflation outbreak of 2021-2022 coincided with an unusual rise in profit shares in some sectors, indicating that businesses were able to increase earnings even as their costs rose. This implies, that companies were raising prices above the level needed to offset higher input costs, with the result that profits contributed substantially to inflation in that period.

Weber and Wasner (2023) argue, that the recent inflation is best understood as a profit-driven or *sellers' inflation* process, as opposed to a classic excess-demand (or wage-driven) intuition. They identify a sequence of stages in this process, according to which it typically unfolds in three stages: (1) a cost shock in upstream sectors (like energy) triggers initial price hikes and windfall profits; (2) downstream firms propagate or amplify the shock by raising their own prices, sometimes increasing markups; (3) workers respond to real wage losses by demanding higher wages, which can perpetuate inflation through a wage-price spiral if the conflict remains unresolved.

This three-stage dynamic-cost shock, markup propagation, and wage response-forms the core of the Post-Keynesian understanding of inflation as a manifestation of conflicting income claims. Unlike aggregate models, which are largely indifferent to sectoral disaggregation and treat sectoral price shocks symmetrically across industries, the Post-Keynesian framework offers a different view. It conceives of inflation as a structurally embedded process shaped by institutional settings and the distribution of market power. Crucially, its microeconomic grounding allows for systematic differentiation across sectors: inflationary dynamics can vary depending on industry characteristics, market structures, and position within the value chain. This perspective thus opens an analytical angle that moves beyond aggregate relationships and permits a more granular, sector-specific understanding of inflation-an approach that will be of central importance in the empirical analysis that follows.

2.3.2 Transmission of Sectoral Shocks in Industrial Networks

Building on the Post-Keynesian framework the work of Acemoglu et al. (2012, 2016) and Gabaix (2011) offer a complementary perspective on how sector-specific disturbances propagate through interconnected production systems. Their contribution, similarly to this thesis, relies on IO analysis. They demonstrate, that shocks originating in one industry, whether supply- or demand-driven, do not remain contained, but instead ripple across the economy via complex IO linkages. While Acemoglu and colleagues primarily focus on quantities, the structural transmission mechanisms identified in their work provide information of network transmission in general. This network-based view adds an additional component to the theoretical foundation for understanding how localized price changes may generate system-wide inflationary effects, aligning closely with the Post-Keynesian emphasis on cost-push dynamics.

A complementary line of argument is provided by Gabaix (2011), who introduces the so-called granular hypothesis. He shows that idiosyncratic shocks to very large firms–especially those occupying central roles in supply chains–can disproportionately influence aggregate fluctuations. In economies with fat-tailed firm size distributions, the standard diversification logic of representative-agent models fails: large firms' output and pricing decisions shape not only sectoral performance but also macroeconomic aggregates.

The most two important theoretical conceptions derived from this line of literature are as follows:

1. **Sectoral Shocks and Directional Propagation:** Acemoglu et al. (2012, p. 1985) and Acemoglu et al. (2016, pp. 274–275) distinguish between downstream and upstream propagation mechanisms. A negative supply-side shock, such as the recent supply-chain disruptions or energy price spikes, in an upstream sector raises input costs for downstream industries, thereby triggering output reductions, and/or price increases in this analysis, along the production chain. Conversely, demand-side shocks (e.g. fiscal stimulus) propagate upstream, increasing the demand for intermediate goods and potentially putting upward pressure on prices and factor utilization in supplier sectors. These propagation dynamics challenge the representative-agent assumptions of aggregate models and highlight the heterogeneity of inflationary effects across sectors. This thesis focuses on upstream propagation.

2. **The Role of Network Structure:** The intensity and reach of shock propagation are fundamentally shaped by the topology of intersectoral linkages. Using the Leontief inverse matrix, Acemoglu et al. (2016) show how both direct and indirect effects can be formally traced. In particular, sectors that are highly connected, either as large suppliers or essential customers, act as shock amplifiers. Their centrality within the production network makes them disproportionately influential in determining aggregate responses. This underscores the asymmetric transmission of price pressures and the importance of sector-specific characteristics, such as market power or the ability to substitute inputs, in shaping inflationary outcomes.

These propositions are highly compatible with the Post-Keynesian view of inflation as a structurally embedded, micro-founded process. When an upstream shock raises costs, firms in downstream sectors may attempt to maintain markups by increasing their prices. From a price perspective, if workers respond by demanding nominal wage increases to protect real incomes, and firms again pass on these higher costs, a wage-profit spiral may ensue. The origin of the shock may be sectoral, but the inflationary consequences unfold across the network, depending on the degree of market power and resistance capacity among claimants. Thus, sectoral shock propagation becomes the microeconomic transmission mechanism through which distributional conflict is activated in a structurally interdependent economy.

Recognizing that inflationary impulses often stem from specific nodes in the production network has two important implications. First, macroeconomic interventions must be complemented by sector-specific tools. For example, targeting energy-intensive industries or monopolistic sectors with regulatory, fiscal, or price control measures. Second, the heterogeneity of sectoral positions and cost structures implies that inflation is not uniformly transmitted. This opens up analytical space to study differentiated inflation patterns across sectors. This space will be explored in the empirical part of this thesis. In contrast to aggregate models that treat sectoral shocks as homogeneous or neutral in structure, the network approach emphasizes economic structure, input dependencies, and pricing power as key transmission channels.

2.4 Micro and Macro Determinants in the Present Analysis

After reviewing inflation theories and how they resolve the tension between micro and macro determinants, the theoretical fit for this thesis should briefly be discussed. As indicated at the beginning of this chapter, this thesis subsumes itself primarily in the Post-Keynesian tradition. Its central insight–that inflation is not merely an aggregate, monetary phenomenon but a structurally embedded process rooted in distributional conflict and firm-level price-setting–offers a compelling micro-foundation for macroeconomic outcomes. This framework explicitly acknowledges the heterogeneity of sectors and agents, and the role of institutional and structural asymmetries in shaping inflationary dynamics. By focusing on cost-push mechanisms, markup behavior, and intersectoral price transmission, Post-Keynesian theory provides a consistent rationale for why price changes in certain industries can disproportionately affect the general price level.

In contrast, classical and monetarist approaches largely treat inflation as a nominal phenomenon driven by excess money supply, while assuming flexible prices and full employment in the long run. Such models offer little room for analyzing sectoral asymmetries or institutional frictions. New Keynesian theories incorporate nominal rigidities and rational expectations, but they continue to rely on representative-agent frameworks that abstract from structural heterogeneity and distributional conflict. As a result, they lack the analytical tools to explain why sector-specific shocks might propagate unevenly across the economy or trigger sustained inflationary dynamics.

Hence, the present thesis builds on a Post-Keynesian foundation by investigating which sectors exert disproportionate influence in the price system–those whose price dynamics are not only relevant locally but have system-wide implications. The theoretical contributions regarding conflict inflation will set the tone for the discussion of results, while the work by Acemoglu and colleagues provides insights into transmission of idiosyncratic shocks. The goal of identifying *systematically significant prices* can help to identify sectors, where prices might increase more than necessary. Also, it can help to guard pressure points in the economic structure of countries in the EU.

The next section will introduce and operationalize the concept of *systemically significant prices*, drawing on recent contributions that combine Post-Keynesian insights with input–output network analysis (Ferreira et al., 2024; Ipsen et al., 2023; Weber et al., 2024). This concept provides an explicit way to bridge micro-level shocks and macro-level inflation, moving beyond traditional aggregate models and towards a disaggregated, structural understanding of inflation transmission. The next section defines this concept in detail.

2.5 Defining Systemically Significant Prices

To translate the preceding theoretical rationale into an operational concept, this thesis adopts and develops the notion of *systemically significant prices*–prices whose sector-specific changes exert disproportionate effects on the general price level. The concept was originally introduced by Hockett and Omarova (2016). The analytical goal is to identify industry-level prices that, when shocked, generate broad inflationary pressures via inter-sectoral IO linkages and household consumption channels. In other words, the aim is to detect sectors whose price dynamics are not merely local perturbations, but potential system-wide inflation triggers.

Following Weber et al. (2024) and Ipsen et al. (2023), based on the foundational work of Hockett and Omarova (2016), the concept draws inspiration from the post-crisis designation of systemically important financial institutions (SIFIs). Just as the failure of certain financial entities can jeopardize system stability, the price dynamics of certain sectors may propagate inflationary disturbances through their structural embeddedness in the economy. Hockett and Omarova (2016, p. 2) argue that prices and indices can function analogously to financial institutions in posing systemic risks, and should therefore be analyzed with similar conceptual tools.

In this framework, systemically significant prices are defined by their potential to influence the aggregate price level, as captured by the Consumer Price Index (CPI). Importantly, systemic significance is not treated as a fixed attribute of any given sector, it can vary of time. For the first part of the analysis, the term *latent* systemically significant prices implies, that the analysis does not rely on observed crisis events, but rather on structural price characteristics–specifically, long-term volatility. This approach is partly motivated by data limitations, as FIGARO consistent industry-level price data are not available for the pandemic years. However, inspired by Weber et al. (2024) and Ipsen et al. (2023), drawing on average sectoral volatility over a longer pre-crisis period (2000-2014), the analysis captures enduring sectoral properties rather than reactions to singular shocks. The resulting stress-test simulates uniform price shocks calibrated to historical volatility, revealing which sectors are structurally prone to transmit inflationary pressures across the economy.

2.5.1 Criteria for Systemic Importance

To identify sectors which pose such risks, systemically significant prices [7], are designated based on their influence across the economy. The key criteria are: (i) their ubiquity in production processes; (ii) the widespread investment exposure to related asset classes; and (iii) their role as benchmarks in other pricing decisions (Hockett & Omarova, 2016, p. 3). As Weber et al. (2024, p. 300) point out, this approach focuses on the first criterion.

Ubiquity of a product or item in an economic system is an important indicator for the significance of its price (Hockett & Omarova, 2016, p. 5). Cheng et al. (2017, p. 111) identify similar criterions, with two of the main indicators are *network connectivity*[8] and *hub centrality*[9]. Interest rates exemplify this. They influence borrowing costs and indirectly shape the pricing of goods, services, and assets. This makes them critical drivers of

[7] In Hockett and Omarova (2016) referred to as *Systemically Important Prices and Indices (SIPIs).*

[8] *Network connectivity* measures how extensively an industry is linked to others in the IO network, indicating its structural pervasiveness across the economy.

[9] *Hub centrality* reflects the extent to which an industry supplies outputs to important downstream industries, amplifying its influence as a systemic transmitter of shocks.

overall inflation and CPI movements. But this approach focuses on goods, where ubiquity is equally well applicable.

Since virtually every product is used to some extent as an input for the production of other goods, they can be more or less central to a range of production processes. If one assumes a price increase in a good, the producers will try to pass these onto the good's price as an input for another industry (or consumers). Thus, costs have increased for the next production process downstream. Again, the increased costs might be transmitted to the next buyer, and so on. If a good is a primary input for many industries, a price increase can cascade through many IO linkages in the supply chain. The likeliness of such centrality is generally increased for upstream inputs, like energy, raw materials, labor, etc. Thus: *The more ubiquitous a good is, the more likely its price is systemically significant.*

The empirical strategy of this analysis opens two additional channels of inflation impact (Weber et al., 2024): (i) the price volatility of a sector and (ii) its share in household consumption.

(i) **Price volatility as a channel:** Sectors with historically high price volatility are more likely to generate inflationary pressure when external shocks occur. Volatility reflects the inherent instability or sensitivity of a sector's pricing to supply disruptions, geopolitical tensions, or speculative dynamics. When a volatile sector also occupies a structurally central role in the production network, its price swings are more likely to cascade through inter-industry linkages, amplifying overall inflationary effects. Thus, volatility serves not only as a risk marker but as a multiplier of potential price disturbances within the economic system. Thus: *The more volatile the prices in a sector, the more likely its price is systemically significant.*

(ii) **Consumption share as a channel:** The weight of a sector in household consumption determines the direct transmission of sectoral price changes into the Consumer Price Index (CPI). Even if a sector is not highly central in IO terms, a large share in consumer expenditure - such as food, housing, or energy - ensures that its price dynamics have an outsized effect on measured inflation. This consumption linkage is especially relevant during overlapping emergencies, where essential goods may experience both supply constraints and demand persistence, making their inflationary effects more persistent and socially salient. Thus: *The higher the share of a sectors output in private household consumption baskets, the more likely its price is systemically significant.*

In sum, prices deemed systemically significant typically fulfill at least one of three structural criteria: (i) ubiquity as a production input, (ii) elevated price volatility, or (iii) a substantial share in final household consumption. These dimensions–input centrality, price instability, and CPI relevance–are jointly integrated into the empirical framework that follows. Importantly, systemic significance is not a static attribute. As production structures, consumption patterns, or energy technologies evolve, the role of specific inputs may change over time. For instance, as Weber et al. (2024, p. 301) note, oil has historically been a paradigmatic example of a systemically significant input, but its role may diminish as economies decarbonize and diversify their energy mix.

The following section outlines the methodological approach used to identify such systemically significant sectors. It introduces the data sources, shock simulation design, and procedures used to estimate price transmission across the input–output network and into consumer prices.

Chapter 3

Methodology

The Leontief price model is a fundamental component of IO (IO) analysis and provides the theoretical framework for analyzing inter-sectoral cost-price relationships within industry networks (Miller & Blair, 2022). It captures the economy as an industry network, rendering it adequate for observing how changes in sectoral input prices propagate throughout the price formation system. This chapter introduces the classical Leontief input–output price model and then presents the modifications and data needed for this thesis's analysis. All computations and simulations were implemented in Python. The full codebase is available at https://github.com/DaBurb/Masterthesis-Burbach-Economics.

3.1 The Classical Leontief Price Model

Introduced by Vassily Leontief (1936), the original model focused on physical quantities, but for this analysis, the price model is used to translate these into monetary values (Miller & Blair, 2022, pp. 42–55). The Leontief price model represents the economy as a system of linear equations linking sectoral input costs to output prices. It is assumed: (i) homogeneous output per sector, (ii) fixed input proportions (no substitution, constant returns to scale), (iii) linear production coefficients, and (iv) exogenous final value-added and demand (Valadkhani & Mitchell, 2002, p. 124). Under these assumptions, each sector's price can be expressed as the cost of its required inputs plus value-added.

Based on these assumptions, each sector's total output $x_j P_j$ can be expressed as the sum of its required intermediate input values from other sectors, expressed through *technical coefficient* a_j multiplied with the quantiative output x_j and the price of that input P_i, plus the *primary inputs* (wages, taxes and subsidies, etc.), which combine into the *Value Added* component v_j, and imports m_j:

$$x_j P_j = x_j a_{1j} P_1 + ... + x_j a_{ij} P_i + ... + x_j a_{nj} P_n + v_j + m_j \tag{3.1}$$

To simplify the model, assume unit output normalization $x_j = 1$, which is valid under constant returns to scale and allows a focus on per-unit pricing (Ipsen et al., 2023; Valadkhani & Mitchell, 2002; Weber et al., 2024). Furthermore, since the data in this thesis is sourced from a global IO database that does not separately track imports, the import term m_j is omitted. The system of inter-industry price equations therefore takes the following form:

$$
\begin{cases}
P_1 = a_{11}P_1 + a_{21}P_2 + \cdots + a_{n1}P_n + v_1 \\
P_2 = a_{12}P_1 + a_{22}P_2 + \cdots + a_{n2}P_n + v_2 \\
\quad \vdots \\
P_n = a_{1n}P_1 + a_{2n}P_2 + \cdots + a_{nn}P_n + v_n
\end{cases}
\tag{3.2}
$$

In matrix form, this IO system is expressed as:

$$P = A'P + v \tag{3.3}$$

Solving for the price vector P yields the classical *Leontief price model*:

$$P = (I - A')^{-1} \cdot v \tag{3.4}$$

The matrix $L = (I - A')^1$, known as the Leontief inverse, captures the output required from each sector to satisfy a unit increase in final demand (Miller & Blair, 2022, pp. 20–21).

Due to the assumption of constant technical coefficients (a_{ij}, P would be a function of v alone, since imports are not included. This equation could therefore be used to assess price impacts from changes in *VA* (Valadkhani & Mitchell, 2002, p. 124). This does not directly allow for the analysis of *industry* level price changes.

3.2 The Modified Leontief Price Model

This thesis applies a modified version of the classical Leontief price model to allow for the sector-specific simulation of price shocks. The approach was initially introduced by Valadkhani and Mitchell (2002) and has been extended in recent contributions by Weber et al. (2024) for the US economy and Ipsen et al. (2023) for the European Union. The central idea is to treat specific sectors as exogenous to the price formation system. This enables a sector-by-sector analysis of how a price shock in one sector propagates through the rest of the economy via IO linkages. According to Miller and Blair (2022, Ch. 14.2), such a treatment constitutes a “mixed model” framework.

In Valadkhani and Mitchell (2002), the price of a specific commodity (e.g., coal, petroleum) is set exogenously, excluding its price and value-added components from the endogenous system. Consequently, the model partitions the economy into one or more exogenous sectors, whose prices are given, and a remaining set of endogenous sectors, whose prices are computed endogenously.

Formally, this yields a partitioned system of the form:

$$\begin{bmatrix} P_X \\ \\ P_E \end{bmatrix} = \begin{bmatrix} A'_{XX} & A'_{EX} \\ \\ A'_{XE} & A'_{EE} \end{bmatrix} \cdot \begin{bmatrix} P_X \\ \\ P_E \end{bmatrix} + \begin{bmatrix} v_X \\ \\ v_E \end{bmatrix} \tag{3.5}$$

where P_X is a $x \times 1$ vector of exogenously set prices (with $x = 1$ in the case of a single exogenous sector), and P_E is an $e \times 1$ vector of endogenous prices (with $e = n - x$). Following the formulation by Weber et al. (2024, p. 319), Valadkhani and Mitchell (2002, p. 125) and Nikiforos et al. (2024, p. 348), the coefficient matrix can be decomposed into four submatrices: A'_{XX} reflects how exogenous sectors source inputs from each other, A'_{EX} captures their reliance on endogenous sectors, A'_{XE} indicates the inputs endogenous sectors acquire from exogenous sources, and a final submatrix A'_{EE} showing the interactions among endogenous sectors themselves. The value-added vectors v_X and v_E represent, respectively, the value-added shares of the exogenous and endogenous sectors, aligned in dimension with P_X and P_E (Valadkhani and Mitchell 2002, p. 125; Weber et al. 2024, p. 319).

Because P_X is exogenously fixed, the system can be reduced to a conditional price equation for the endogenous sectors:

$$P_E = A'_{XE} P_X + A'_{EE} P_E + v_E \tag{3.6}$$

Solving for P_E yields:

$$P_E = (I - A'_{EE})^{-1} A'_{XE} P_X + (I - A'_{EE})^{-1} v_E \tag{3.7}$$

In this framework, v_E is assumed constant. Thus, starting from equation (2.7), changes in the endogenous price vector ΔP_E are determined solely by the change in exogenous prices ΔP_X and the structure of the IO system:

$$\Delta P_E = (I - A'_{EE})^{-1} A'_{XE} \Delta P_X \tag{3.8}$$

This final expression is the core mechanism used in this thesis to simulate the propagation of price shocks. It quantifies the extent to which an exogenous sectoral price increase transmits to downstream industries through production interlinkages. The Leontief inverse $(I - A'_{EE})^{-1}$ amplifies and redistributes the initial shock, depending on the structure of the endogenous production network.

3.3 Empirical Application

The empirical application of the modified Leontief price model requires the integration of three key components: (i) inter-industry IO data, (ii) the specification of sectoral price shock amplitudes, and (iii) a synthetic consumer price index (CPI) for aggregating and interpreting sectoral impacts on the macroeconomic price level.

The underlying data are sourced from the FIGARO industry-by-industry IO tables provided by Eurostat (Eurostat, 2024a). These tables provide harmonized, multi-country production structures for the EU and its member states. However, the FIGARO dataset does not include corresponding producer price data. As a result, it is not possible to directly simulate actual historical shocks, such as those observed during the COVID-19 period.

To address this limitation, the analysis adopts the concept of *latent systemic significance*, as introduced by Weber et al. (2024). Instead of imposing realized price shocks, a stylized stress-test approach is employed, using historical price volatility as a proxy for sectoral risk exposure. This synthetic shock structure enables the simulation of plausible, yet generalizable, price propagation patterns.

To quantify the effect of sector-specific shocks on the overall price level, a synthetic CPI is constructed based on final household consumption weights (Ipsen et al. 2023, pp. 11–13; Valadkhani and Mitchell 2002, p. 126; Weber et al. 2024, pp. 304–305). This allows for the decomposition of direct and indirect contributions to consumer inflation within the inter-industry production network.

3.3.1 Stress Test Scenario

In the absence of complementary producer price indices in the FIGARO data, stress-test scenarios use historical price volatility from the World Input Output Database (WIOD), specifically the Socioeconomic Accounts (SEA) dataset (Timmer et al. 2015; Gouma et al. 2018). Following the methodology of Weber et al. (2024) and Ipsen et al. (2023), the standard deviation of annual producer price changes is computed for each sector over a reference period. These volatilities are treated as normalized sectoral shock amplitudes. Formally, the volatilities are defined as:

$$\sigma^x_{t_0,t_1} = \sqrt{\frac{1}{T}\sum_{t=t_0}^{t_1}\left(\Delta P^x_t - \Delta \bar{P}^x_{t_0,t_1}\right)^2} \tag{3.9}$$

The period T reaches from $t_0 = 2000$ to $t_1 = 2014$, e. g. 15 observations. The yearly percentage price change is denoted $\Delta P^x_t = (P_t - P_{t-1})/P_{t-1}$ and $\Delta P^{\overline{x}}_{t_0,t_1}$ expresses the average price change per year over T

The rationale for this approach is to approximate the sectors' intrinsic systemic relevance based on the volatility of their historical pricing behavior. High-volatility sectors are assumed to pose a greater risk of inflationary propagation when shocked. Each stress test is conducted by setting the respective sector's price exogenously, increasing it by the calculated volatility, and calculating the resulting changes in all endogenous prices using the modified Leontief model. The standard deviation approach captures the magnitude and frequency of deviations from the average price change, making it a superior fit for this approach, compared to, for example, the average. It reflects the instability and unpredictability of a sector's pricing behavior, e. g. the price shock potential. This procedure is repeated for all sectors in a one-at-a-time fashion, allowing the identification of those sectors whose price changes exert the greatest system-wide influence. In contradition to Weber et al. (2024), but in accordance with Ipsen et al. (2023), prices for specific commodities (Housing, Oil and gas extraction, etc.) are not exogenously determined in the model. Weber et al. (2024) argue, that the logic of price determination behaves different for certain types of commodities. The present analysis ignores this, in favor of treating each sector exactly the same to yield comparable results. Additionally, data limitations play a role.

3.3.2 Synthetic Consumer Price Index

To translate sectoral price changes into consumer price effects, a synthetic CPI is constructed (Ipsen et al., 2023; Valadkhani & Mitchell, 2002; Weber et al., 2024). This index weights each sector's price according to its share in final household consumption, as recorded in the FIGARO final demand block[10]. The CPI impact of a given sector is calculated as the weighted sum of price changes across all sectors:

$$\Delta \mathrm{CPI} = \sum_{i=1}^{n} w_i \cdot \Delta P_i \tag{3.10}$$

where w_i denotes the consumption weight of sector i and ΔP_i the change in its price following the simulated shock. Based on Ipsen et al. (2023, p. 12), the CPI can further be used to estimate the impacts on different regional aggregates, as explained in appendix A.3. Due to the focus on the EU28, it will be included as one of the possible weights. Equation (2.10) will be calculated for the chosen regional samples. This enables a consistent comparison across sectors and regions and especially the decomposition in direct and indirect inflation impacts, which will be introduced in the next section.

3.3.3 Propagation and Decomposition

Once the price system is perturbed by a sectoral shock, and with the synthetic CPI weights at hand, the resulting change in prices across the economy can be decomposed into two components: (i) direct effects stemming from the shocked sector's consumption weight, and (ii) indirect effects that materialize through inter-sectoral IO linkages. Adding both yields the total inflation impact. If *inflation impact* is used in this thesis, it refers to the weighted impact of the endogenous price volatility effect. The synthetic CPI is the

[10] P3_S14 – Final consumption expenditure of households

third influential pillar regarding the overall influence of and industry on inflation. Even though some sectors might be less impactful on other industries, due to its high share in personal demand it can be systemically significant. Vice versa, some sectors might not exert direct inflationary pressure, but produce inputs for sectors with strong significance for final demand, increasing their indirect impact.

The decomposition is structured as follows:

$$IP_{\text{dir}} = c_x \Delta P_X, \tag{3.11}$$

$$IP_{\text{ind}} = \sum_{i \neq x} c_i \Delta P_E^i, \tag{3.12}$$

$$IP_{\text{tot}} = \sum_{i \neq x} c_i \Delta P_E^i + c_x \Delta P_X. \tag{3.13}$$

where j denotes the shocked sector. This decomposition is crucial to understanding the systemic role of sectors that, despite having low direct CPI weights, significantly affect inflation through production networks. Results are reported both in absolute terms and as percentages of total CPI effects.

3.4 Gas Price Shock Analysis

The second part investigates the inflationary effects of a sector-specific price shock: the increase in natural gas prices. In contrast to the latent systemic stress tests described above, this module introduces an exogenous shock into the price system, which is derived from a real world scenario, and quantifies its propagation through the European production network. The real world scenario in question is the price shock for natural, caused by sanctions on Russia due to the invasion into Ukraine in 2021. Thus, the inter-industry network of 2021 will be shocked. The methodology largely follows the structure of the modified Leontief price model outlined in Section 3.2, with adaptations to reflect the empirical nature of the shock scenario and the specificity of the gas-related sectors.

3.4.1 Shock Specification and Sectoral Disaggregation

Gas price shocks are modeled as exogenous increases in the price of the natural gas extraction sector. However, the official FIGARO classification aggregates this sector under a broad mining and quarrying category (B), which includes non-energy-related activities. To address this limitation, this thesis makes use of the higher sectoral granularity provided by the EXIOBASE3 IO tables (Stadler et al., 2018). Specifically, the EXIOBASE3 dataset allows for the identification and separation of natural gas extraction activities[11], enabling a re-allocation of shock values to a more targeted sector within FIGARO.

To achieve consistency, EXIOBASE3 and FIGARO tables are harmonized via a proportional splitting procedure, which will be explained in more detail in the data section. The value of intermediate flows and outputs in FIGARO's parent sector B is adjusted using EXIOBASE3's relative shares of natural gas extration and related activities within the broader mining and quarrying activities. This reweighting yields a more precise impact estimation of a gas-specific price shock.

3.4.2 Shock Calibration and Application

The amplitude of the natural gas price shock is calibrated to the 500 % one-off increase used by Cucignatto et al. (2023), who ground their scenario in market data from the Title Transfer Facility (TTF) gas-price index. The TTF is a virtual trading point for natural gas, based in the Netherlands, and regularly serves as a European benchmark for wholesale natural gas prices. Its spot and forward curves capture extreme volatility during the 2021–2022 period following the Russian invasion of Ukraine. By imposing the 500 % increase at the extraction node, *i. e.* treating the *Natural Gas Extraction* sectors as fully exogenous, the model traces the maximum downstream propagation of price shocks through the entire production network.

Operationally, the shock enters the model through an additive exogenous price vector

$$\Delta P_X, \qquad (\Delta P_X)_{c,\mathrm{B\_gas}} = \begin{cases} 5 & \text{if the gas row is } \textit{retained} \text{ in the scenario,} \\ 0 & \text{otherwise,} \end{cases}$$

so that every retained gas-import row experiences a +500% price change while domestic extraction remains at 0.

Two binary masks M are used to zero selected columns in the mixed coefficient block A_{XE} (all notation as in Eq. (**??**)):

(i) *Extra-EU import shock:* set $M_{rs} = 0$ for every gas flow that is domestic **or** intra-EU; only third-country imports transmit the shock.

(ii) *Full-import shock:* set $M_{rs} = 0$ only for domestic gas flows; both extra-EU and intra-EU imports transmit the shock.

The masked exogenous coefficients matrix $A^*_{XE} = M \odot A_{XE}$ are inserted into the

[11] Two categories are used to account for natural gas extraction in the supply chain: *Extraction of natural gas and services related to natural gas extraction, excluding surveying* and *Extraction, liquefaction, and regasification of other petroleum and gaseous materials.* The latter refers to Liquefaction, e. g. the conversion of natural gas into a liquid form (LNG), and the regasification, e. g. conversion of LNG back into gaseous state for distribution and use.

transposed Leontief price model

$$\Delta P_E = (I - A'_{EE})^{-1} \, (A^*_{XE})' \, \Delta P_X,$$

after which sectoral results are aggregated with CPI weights (Section 2.6.3). Because the experiment is conducted in *changes*, setting matrix entries to 0 is innocuous: a nulled coefficient cannot transmit any part of the shock, while it also does not affect endogenous transmission (for a detailed explanation, see Appendix A.4).

The two scenario runs therefore differ *only* in the definition of M, allowing a clean decomposition of the pure intra-EU transmission channel.

The remainder of the procedure follows modified Leontief framework introduced in Section 2.4. After the scenario mask is applied, the endogenous block A_{EE} remains untouched, whereas the exogenous block is replaced by its respective masked counterpart $A^*_{XE} = M \odot A_{XE}$. This is the *only* modification; the model that is subsequently solved

$$\Delta P_E \; = \; \big(I - A'_{EE}\big)^{-1} \big(A^*_{XE}\big)' \, \Delta P_X$$

is otherwise identical to the baseline specification. All downstream steps, aggregation to sector or country level and application of CPI weights, albeit not distinguishing direct and indirect effects, proceed unchanged, ensuring full comparability with the standard implementation discussed earlier.

3.5 Data Sources and Preparation

Some of the novelties presented in this thesis are based on the underlying data used. During the process of writing this thesis, the paper by Ferreira et al. (2024) was published, which carried out an IO analysis looking for systemically significant prices based on the FIGARO dataset. The difference lies in the applied stress test scenarios. To the authors knowledge, this is the first attempt to adapt FIGARO IO database to two other databases, WIOD (Timmer et al., 2015) and EXIOBASE3 (Stadler et al., 2018, 2021). This facilitates several aspects of this thesis. The first one is the usage of complementary price data from WIOD which is provided in full coverage over all countries and years. Second, EXIOBASE3 allows to split up extraction of natural gas from other mining, quarrying and extraction of fossil energy carriers.

3.5.1 FIGARO Input Output Database

The FIGARO tables, provided by Eurostat, form the primary empirical basis for the structural IO analysis in this thesis (Eurostat, 2024a, 2024b). They contain harmonized inter-country IO tables for 27 EU Member States, 18 additional trading partners, and a rest-of-the-world aggregate, resulting in a total of 36 regions. FIGARO follows the NACE Rev.2.1 classification, offering data for 64 industries, and is valued at basic prices [12]. Coverage spans the years 2010 to 2022, allowing for thirteen annual observations. The dataset includes full final demand and value-added components. It must be pointed out, that FIGARO consists of global IO tables, as it was already pointed out, where imports are not explicitly denoted. Overall, the basic IO tables provided by FIGARO are well suited to inflation-related studies in the European production network.

The key advantage of FIGARO over alternative sources, such as WIOD, lies in its recency, and its EU focus, which seems optimal for this analysis. In particular, the dataset includes the years immediately preceding the COVID-19 crisis, while excluding the crisis itself, as the base year for the analysis is set to 2019. This ensures that the sectoral network structure is contemporary, but not distorted by the exceptional price volatility observed during the pandemic. While 2019 is the year focused for the analysis of latent systemically significant prices, the analysis will be conducted for each year. This might reveal interesting dynamics in the industrial network over the observed time period. A major limitation of FIGARO, however, is the absence of complementary price data. They do not include sector-specific inter-industry price indices. To overcome this, FIGARO is adapted to the WIOD price system, which contains this vital information. The recent study by Ferreira et al. (2024) uses price data by EUKLEMS, which has certain limitations regarding coverage. The decision for the WIOD adaption was made, to achieve full and consistent coverage over all sectors and years for a long period.

3.5.2 World Input Output Database

The World IO Database (WIOD) complements FIGARO by providing long-run sector-specific price information (Timmer et al., 2015). WIOD contains data for 56 industries based on the ISIC Rev. 3 classification, covering 27 EU countries and 15 trading partners over the period from 2000 to 2014. Its socioeconomic accounts complementary dataset includes inter-industry price indices (II_PI), which allow for the construction of price volatility estimates used to simulate stress-test scenarios. Although WIOD lacks more recent years and has slightly lower structural resolution than FIGARO, its extended temporal coverage makes it valuable for deriving consistently covered historical price volatilities.

In order to harmonize both databases, several adaptations are undertaken. At the sectoral level, selected NACE Rev. 2 service sectors in FIGARO are aggregated to ensure consistency with WIOD. Specifically, all information and communication activities are grouped under a single composite sector labeled as J; professional, scientific, and administrative services are combined as M_N; health and social work sectors are merged under Q; and recreational, cultural, personal, and membership services are aggregated

[12] For a breakdown of NACE codes with descriptions, and the aggregation for either part of the analyis, consult Appendix A.2

into *R_S*.[13] At the country level, price data are not available for Saudi Arabia, South Africa, Argentina, and the 'Rest of the World' group. These regions are nonetheless retained in the structural model to allow for endogenous propagation effects, even though they are not explicitly subjected to price shocks.[14]

3.5.3 EXIOBASE3 Input Output Database

One methodological innovation of this thesis lies in the targeted disaggregation of the natural gas extraction sector, which is not separately identified in the FIGARO IO tables. FIGARO's official classification aggregates gas extraction under the broad mining and quarrying sector (B), which includes non-energy-related industries such as metal ores and other mineral extraction. This aggregation poses a challenge for accurately modeling natural gas price shocks.

To overcome this limitation, data from the EXIOBASE3 IO database is used to disaggregate sector B in FIGARO into *B_gas*, containing only natural gas related activities, and *B_nongas*, which contains other mining and extraction activities. EXIOBASE3, laying focus on environmental extensions, provides more granular sectoral detail and distinguishes between different types of fossil energy extraction, including the necessary dedicated sector for natural gas. Using this finer resolution, relative shares of natural gas extraction in total mining output are computed for each country, sector, and year in EXIOBASE3. These shares are then used to reweigh the intermediate flows in FIGARO's sector B, thereby estimating the share attributable to gas extraction.

Formally, let $g_{i,c}^{\text{EXIO}}$ denote the share of gas extraction in the total mining output for country c and year i in EXIOBASE3. These shares are applied to the FIGARO matrices (Z and X) to construct a synthetic gas sector embedded within FIGARO's structure:

$$Z_{B,\cdot}^{\text{gas, FIGARO}} = g_{i,c}^{\text{EXIO}} \cdot Z_{B,\cdot}^{\text{FIGARO}}, \quad X_{B}^{\text{gas, FIGARO}} = g_{i,c}^{\text{EXIO}} \cdot X_{B}^{\text{FIGARO}}. \tag{3.14}$$

This harmonization procedure ensures internal consistency and enables the simulation of gas-specific price shocks within the broader FIGARO IO framework. It retains the macro-structural accuracy of FIGARO while incorporating the sectoral detail from EXIOBASE3, which is essential for evaluating energy-driven inflation dynamics in the European production network.

[13]The following NACE codes were aggregated: J = J58, J59_60, J61, J62_63; *M_N* = M69_70, M71, M72, M73, M74_75, N77, N78, N79, N80T82; Q = Q86, Q87_88; *R_S* = R90T92, R93, S94, S95, S96.

[14]NACE code *FIGW1* refers to world regions for which inter-industry price data (II_PI) are not provided in WIOD.

3.6 Limitations and Assumptions of the Methodology

The empirical approach allows a wholistic analysis of economic structures and their price transmission channels. To facilitate this, considerable simplification, aggregation, and underlying assumptions are necessary. These pose important limitations for the analysis, which should briefly be listed here.

The first limitation of input–output analysis is (i) the assumption of fixed technical coefficients. This implies that production recipes remain unchanged regardless of relative price shifts, thereby neglecting firms' ability to substitute inputs when relative costs change. Potential adjustments in input mixes (wages included) are, therefore, not mapped. This can lead to potential overestimation, especially mid- to long-term (Ferreira et al., 2024; Ipsen et al., 2023).

Another key limitation lies in (ii) the lack of separate data on prices and quantities. Input–output tables report only total transaction values, i.e., price times quantity, without disaggregating these components. Consequently, normalization procedures—typically assuming a uniform initial price of one across sectors—are required. This introduces a further layer of abstraction that may distort the measurement of real-world price dynamics (Ferreira et al., 2024; Weber et al., 2024).

(iii) The model also assumes full cost pass-through without accounting for substitution or price sink effects. In reality, firms may partially absorb input cost increases or raise their own prices disproportionately due to market power. Ignoring such heterogeneity in pricing behavior may bias the estimation of sectoral inflation contributions (Ipsen et al., 2023).

(iv) A further constraint is the model's short-term focus. Being a static framework, the Leontief price model abstracts from dynamic adjustments over time, such as technological change, capital deepening, or shifts in consumer demand. As such, it does not capture structural adaptation processes that may significantly alter inflation trajectories in the long run (Weber et al., 2024).

(v) The treatment of income distribution is likewise simplified. The model typically assumes constant nominal wages and profits, thereby overlooking potential feedback loops such as wage-price spirals or profit-driven markup adjustments. These channels may either amplify or dampen inflationary dynamics, depending on the institutional context (Ferreira et al., 2024; Weber et al., 2024).

(vi) Commodity markets represent another blind spot. The model treats these as exogenous, despite their high volatility and sensitivity to global speculation, inventory dynamics, and geopolitical shocks. This is particularly relevant for energy and food sectors, where market behavior often diverges from cost-based pricing logic (Weber et al., 2024).

(vii) Aggregation bias also poses a challenge. The high level of sectoral and regional aggregation inherent in input–output data may obscure heterogeneities—such as differences in supply chain exposure, regional consumption patterns, or the incidence of shocks across income groups. The assumption of a representative consumer further limits distributional analysis (Ipsen et al., 2023).

(viii) Finally, the model focuses exclusively on cost-push mechanisms, neglecting monetary and demand-side drivers of inflation. It does not integrate expectations formation, credit constraints, or central bank responses, which play a central role in macroeconomic inflation theory and forecasting (Przybyliński & Gorzałczyński, 2022).

These limitations underscore that while the Leontief price model can provide valuable structural insights, especially for identifying sectoral transmission channels, its results should not be interpreted as precise forecasts. Rather, they highlight potential points of vulnerability within the economic network, which can inform—but not replace—more dynamic, expectation-driven, and behaviorally rich policy models.

Chapter 4

Analysis

This chapter presents the empirical analysis that operationalizes the theoretical and methodological framework developed in the preceding sections. Building on the concept of systemically significant prices, it investigates how sector-specific price shocks propagate through the European production system and contribute to inflationary pressures at the macroeconomic level. The analysis proceeds in two parts: first, a structural identification of sectors with latent systemic significance across the EU28, based on volatility-adjusted stress tests; and second, a focused simulation of the 2021 natural gas price shock, examining its transmission through national and sectoral channels. Together, these components provide a disaggregated perspective on inflation dynamics, revealing the structural vulnerabilities and transmission mechanisms that conventional aggregate models often overlook.

4.1 Systemically Significant Prices in the EU28

This section identifies sectors whose price changes exert disproportionate influence on inflation outcomes within the EU28 economy. Based on the empirical strategy, the following subsections examine systemic significance from three interrelated perspectives: sectoral composition, geographical structure, and time evolution.

4.1.1 Sector-Level Analysis

The Heatmap in Figure 2 visually allows a first impression on the structure of EU28 price systems. While the overall effect is "hidden" due to the logarithmic scaling, the structure within the network becomes visible. Six sectors immediately stand out: The agricultural sector "Crop and animal production (...)" (A01), "Manufacture of food products, beverages and tobacco products" (C10-C12), "Manufacture of coke and refined petroleum products" (C19), "Electricity, gas, steam and air conditioning supply" (D35), and "Real estate activities" (L68). There is another group of sectors, the "Wholesale (...)" and "Retail (...)" (G45; G46; G47) sectors, which indicate consistent contributions on EU28 CPI inflation over all countries. Intuititvely, these sectors seem to be either upstream input products with a high degree of ubiquity or downstream goods relevant for final consumption. The heatmap also indicates a pronounced impact by about a third of EU28 countries to the right with GB being followed by FR and DE. European core countries seem to exert higher influence on EU28 inflation.

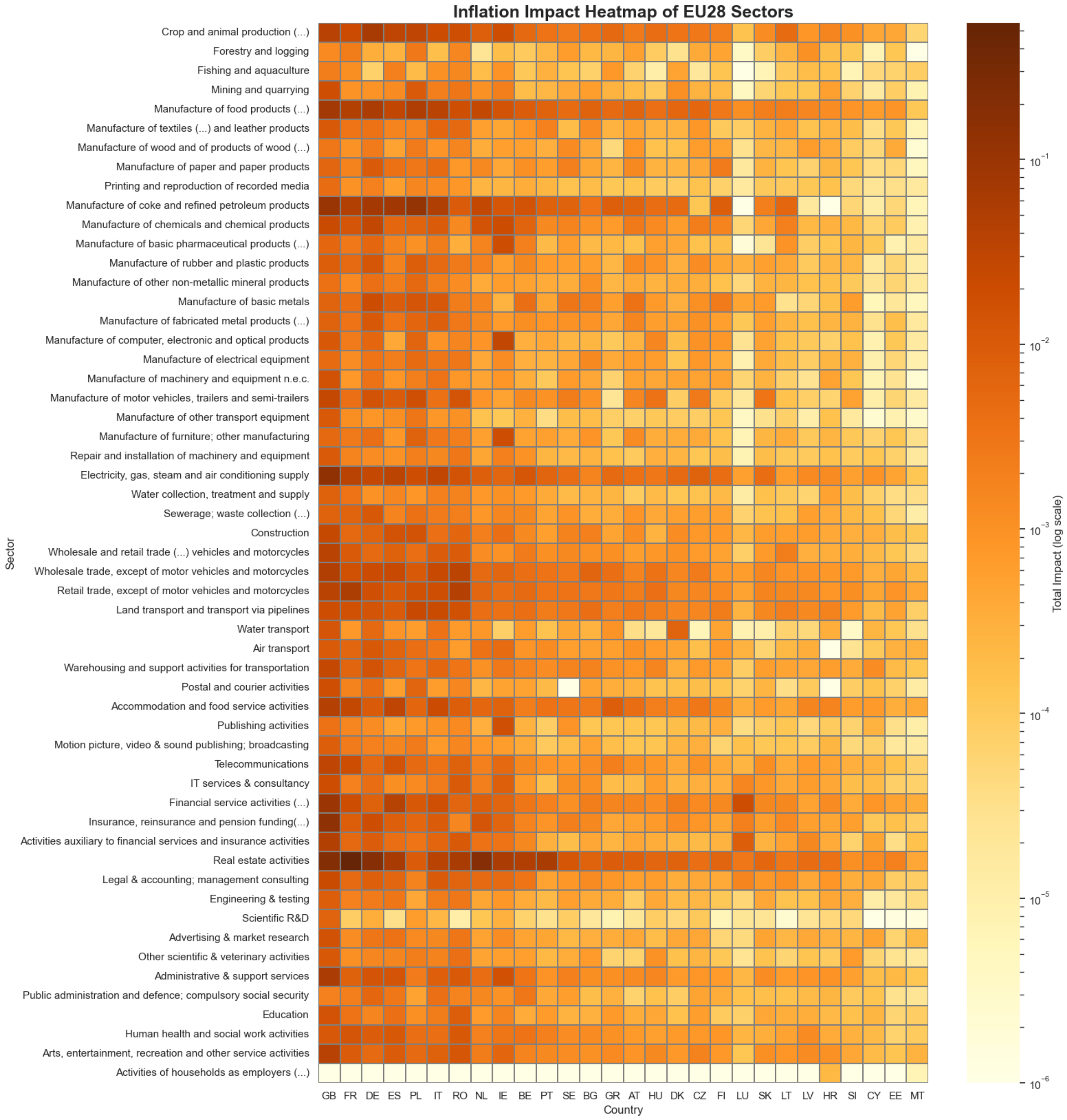


Figure 2: ***Sector-Country Heatmap of CPI-Weighted Inflation Impacts in the EU28 (log scale).*** *This heatmap displays the CPI-weighted inflation impact of sectoral price shocks across EU28 countries on the inter-industry network of 2019. Each cell represents the total inflation impact (direct and indirect) of a sector in a given country, calculated using the Leontief price model and weighted by EU28 household expenditure shares. The color scale is logarithmic, allowing for visual differentiation across several orders of magnitude, from negligible to highly systemic contributions. Columns (countries) are sorted by total impact to highlight national exposure patterns, and increase readability for sectoral impacts. (Own illustration.)*

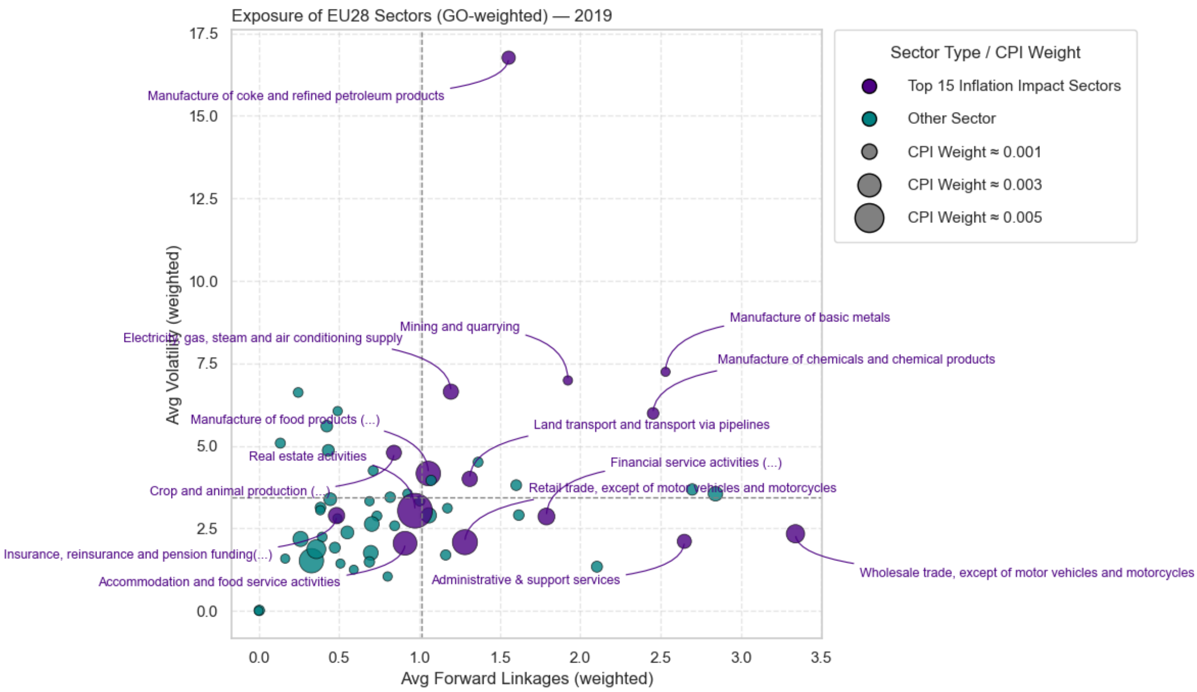


Figure 3: ***Systemic Exposure of EU28 Sectors (2019).*** *This scatterplot displays the average gross output-weighted price volatility and forward linkages of EU28 sectors. Marker size reflects household CPI weight, while colors distinguish the top 15 inflation-impact sectors (purple) from all others (teal). The figure illustrates how inflation risk emerges both from volatile, upstream sectors (top-right quadrant) and from consumption-relevant sectors with high CPI weights (large markers in lower quadrants). (Own illusration, based on Weber et al. (2024)*

A more detailed perspective on sectoral properties is provided in Figure 3, which maps EU28 sectors by their average forward linkages (i.e., their IO centrality) and price volatility, both weighted by respective gross output to reflect sector size. The size of the dot represents household CPI weights, and color indicates whether the sector belongs to the top 15 contributors to aggregate inflation impact (average). The plot reveals a clear bifurcation: sectors with high volatility and/or strong forward linkages, for instance manufacture of coke and refined petroleum products or wholesale trade, occupy the two right quadrants, suggesting a structural position conducive to inflation propagation. Meanwhile, sectors like food production and real estate activities, while less central in production networks, possess relatively high CPI weights, reinforcing their direct impact on consumer price levels. Interestingly, the upper-left quadrant only hosts one top-15 sector, with crop and animal production. Sectors, which are "only" characterized by volatility, do not seem to qualify as systemically significant. The two groups pointed out–volatile upstream sectors and high-weight consumer sectors–coincide with those identified in the heatmap as systematically significant, thereby supporting the dual-channel view of inflation risk: one driven by supply-chain amplification, the other by final demand relevance.

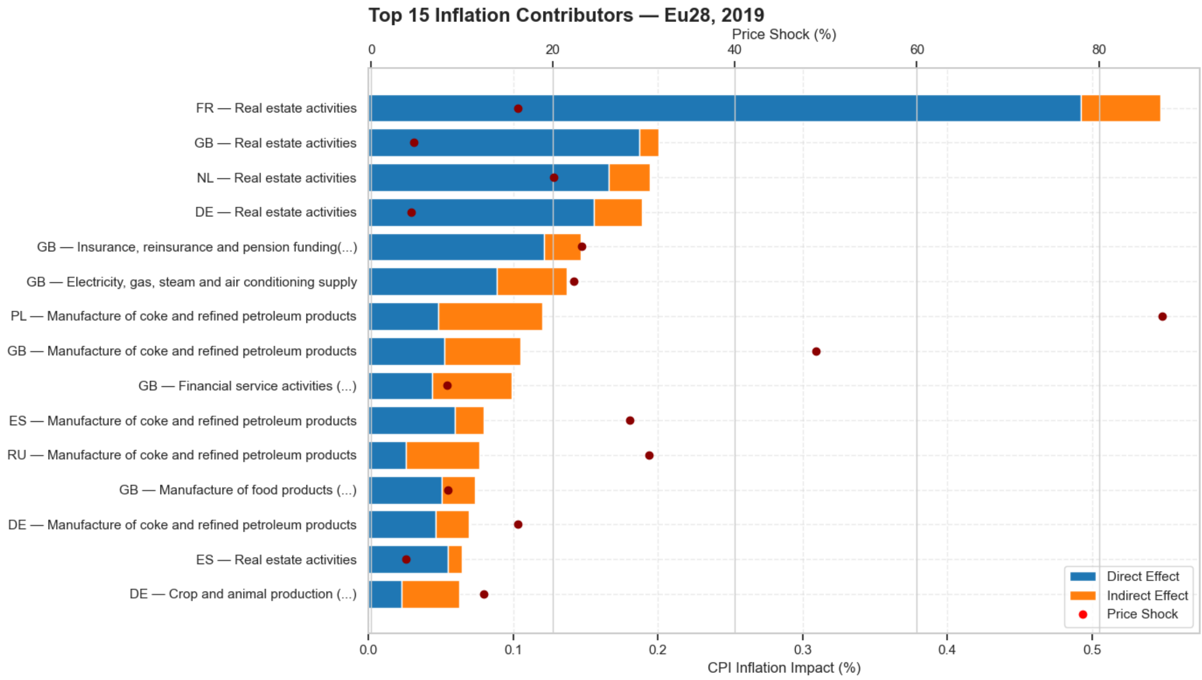
How do the visual impressions of Figure 2 hold up, when zooming into the inflation impact of country and sector combinations? Figure 4a shows the highest impact combinations for EU28 consumer inflation[15]. It reads as follows: Each combination consists of two components, the direct and indirect effect. The dot indicates the price shock, based on the price volatility which was calculated as described in the methodology. Thus, the reader can identify, which type of transmission is prevalent for a sector. The full bar consisting of two components indicates the full estimated effect of a sectoral shock on the EU28 (synthetic) CPI, under the assumptions and limitations laid out in the empirical strategy. Accordingly, the French real estate sector asserts a total impact of $\approx 0.55\%$. Close to 90%, e. g. 0.49 percentage points, of the total impact are asserted directly, compared to around 10%, or 0.055 percentage points.

Generally, six systemically significant clusters can be identified:

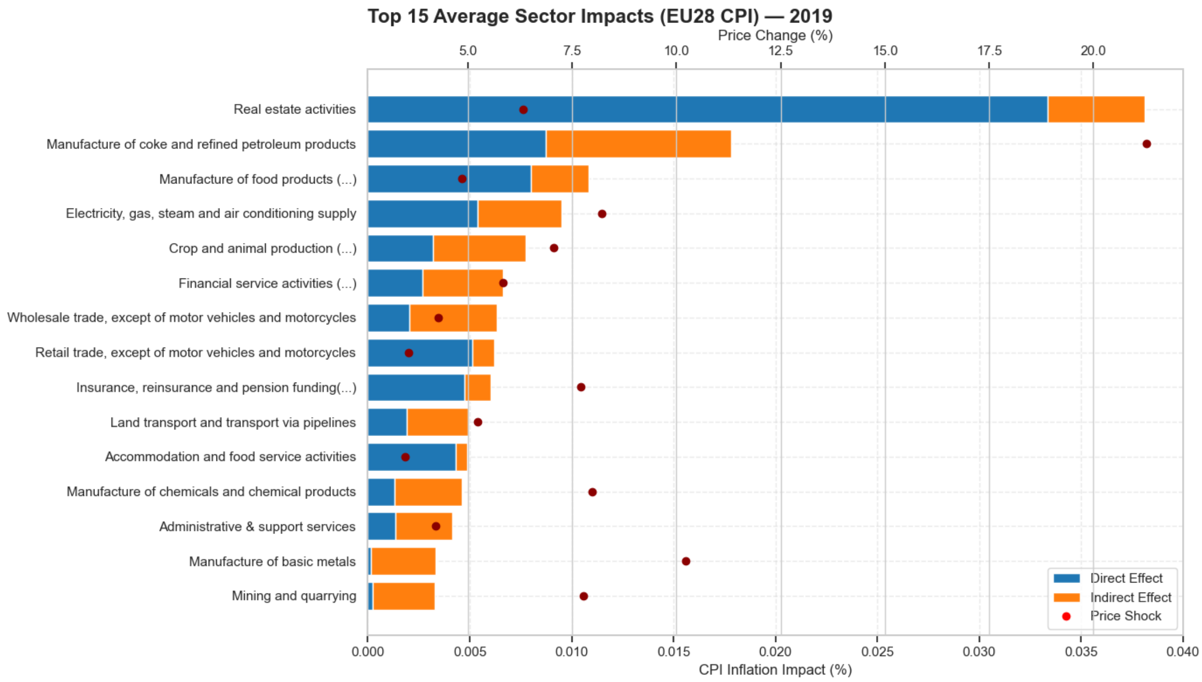
Figure 5 shows that the four sectors with the highest overall inflation impact are all *Real estate activities* located in core EU countries, with the Spanish real estate sector also present among the top 15. Their impact composition is heavily skewed toward direct effects, which aligns with expectations: real estate, renting, and housing constitute the single largest component in most household consumption baskets (Ipsen et al., 2023; Weber et al., 2024). Thus, the systemic significance of prices in the real estate sector arises primarily from their substantial CPI weight. They exhibit relatively modest shock scenarios compared to more upstream industries. This is consistent with the nature of real estate contracts, which are often long-term and price-stable relative to highly volatile commodity markets (Hein, 2024). More interestingly, the indirect effects are not negligible. When ranked by indirect impact, France ranks third, with German and Dutch real estate sectors also among the top 15. Regarding the distribution among different countries' real estate industries, it is heavily right skewed, while also being widely dispersed. The median sector actually ranks second, after manufacture of coke and petroleum. But real estate has some heavy outliers, as it is visible in figure 4a. Thus, while there is maintained relevance over all countries, some countries real estate sectors project outstanding impacts on inflation. The source of the systemic significance is thus threefold: high weights in household consumption, consistent relevance with heavy outliers, and non-trivial forward linkages in the production system—since real estate is relevant not only for consumers, but also for firms through office space, manufacturing sites, and related costs.

The second relevant industrial cluster are Energy-related industries, particularly *Manufacture of coke and refined petroleum products* (C19) and *Electricity, gas, steam, and air conditioning supply* (D35), rank among the most impactful sectors. C19 exhibits a substantial share of indirect effects, as indicated in figure 4b, underlining its role as a foundational input across multiple downstream sectors. Figure 5 highlights C19's high

[15] The full weighted result tables for both parts of the analysis are available here: https://www.dropbox.com/scl/fo/978aibkjgo91ya2tipxmo/AG62H44hQplnc_0YedQDTfg?rlkey=wnbrfbskfoux0or4b30sesw1z&st=hbl28uhw&dl=0



(a) ***Top-15 Sectors According to Inflation Impact on the EU28 Synthetic CPI (2019)***



(b) ***Top-15 Sectors According to Average Inflation Impact on the EU28 Synthetic CPI (2019)***

Figure 4: ***Inflation impact for country-specific and average industries.*** *The full length of each bar shows the estimated overall change in prices from 2019–2020, given the shock scenario, for a specific national industry (a). The country and sector combination is the origin of the respective price change. The yellow component reflects the direct impact on inflation, the violet component the indirect effect. The light blue dot represents the shock scenario, which is constructed with the average price volatility from 2000–2014 (WIOD SEA data). Figure (b) shows the average impacts over all countries in the sample on EU28 synthetic CPI. (Own illustration based on Weber et al. (2024).)*

dispersion and right-skewed distribution, reflecting the exposure of EU28 to highly volatile fuel prices. Also, in combination with the heatmap, some countries have little to no manufacturing capacities of fossil fuels, while other countries like Poland have highly relevant industries. The median sector has a higher impact compared to its real estate counterpart. The high volatility, combined with its inherent ubiquity, elevates C19 to a latent systemically significant sector, with a strong country dependent component (Hein, 2024; Weber et al., 2024). Regarding those country patterns, among the top-15 ranking, the Polish and British sectors are up front, displaying endogenously driven effects with only the Russian counterpart showing a stronger endogenous component. Spanish and German fossil fuel manufacturing seems to have a stronger direct impact.

Thirdly, *Electricity, gas, steam (...)* is more evenly composed across impact channels. Its systemic importance does not stem from a single dominant factor, but from its consistent above average presence across all dimensions: CPI relevance, ubiquity as an industrial input, and volatility due to its reliance on fossil fuels. The combination is displayed in the exposure plot, figure 3. The boxplot confirms this robustness across EU countries, with D35 showing moderate dispersion and consistently high positioning. The IQR is almost evenly distributed around the median, indicating

Food-related industries form a fourth cluster of impactful sectors. The *Manufacture of food products, beverages, and tobacco* (C10–C12) is prevalent here, with *Crop and animal production* (A01) following. Together they represent both ends of the food supply chain. The former, being closer to final consumption, secures a high CPI weight, ranking fourth overall in the synthetic EU28 CPI, comprising 7.9% of household expenditure (Ipsen et al., 2023). In contrast, A01 is positioned upstream, and while moderately volatile, its systemic importance stems largely from its role as a key input into food manufacturing. This IO linkage is evident in Figure 4b: A01's inflation impact is driven more by endogenous transmission than by direct CPI weight. The distribution shows A01 has consistent relevance across countries—with the German agricultural sector the only country-sector to enter the top 15. This broad-based impact rather than country-specific spikes confirms A01's systemic character. The food manufacturing sector shares this consistency but the whiskers and the IQR display pronounced outliers and dispersion, indicating some countries assert strong inflationary pressure in this area compared to others. The high average price volatility, boxplot's wide interquartile range and elevated median position reinforce the latent systemic significance of both sectors in inflation propagation.

Retail and wholesale trade sectors (*G45–G47* and *G46*) form a fourth group. According to figure 4b, their impact composition reflects their position in the supply chain: wholesale is highly central and projects inflation impacts predominantly through endogenous channels, while retail contributes mostly via direct effects tied to its importance in household consumption. Though neither sector features country-level entries in the top 15, their distributions in Figure 5 show robust impact across the EU. The interquartile range for wholesale is particularly broad, highlighting its ubiquity. Retail, while narrower in spread, remains significant due to consistent direct CPI exposure. Notably, the upper quartile (Q3) is distinctly elevated for the prior sector, thereby reaching the mean. Wholesale appears to be only mildly skewed to other sectors surpassing it in average impact, thus, displaying consistently relevant impact from all countries.

The sixth cluster identified are financial and insurance related activities (*K64 and K65).* They show interesting properties compared to the other clusters, especially the latter: There impact is driven by a handful of outliers, in other words, a small group of countries contain highly impactful financial and insurance related industries, while others contribute next to nothing. This might be the result of the sampling over all countries, insurance companies often operate nationally. The heatmap in figure 2 indicates financial hubs, like GB, Spain, and Luxembourg, as well as contributors in the insurance sector with GB, Germany, Italy and the Netherlands. GB has a leading role in all financial and insurance related sectors, ranking 9th and 5th, respectively. Considering the average composition of direct and indirect impacts in Figure 4b, financial activities affects inflation primarily through indirect channels, while the opposite holds for insurance related activities. This, again, reflects their supply chain and final demand characteristics. Figure 3 shows a high degree of embeddedness of financial activities, reflecting the ubiquity of accounting and consulting services in the production network. The systematic significance of the insurance industry is not clearly visible in the scatter plot, which does probably underpin the highly asymmetric relevance across countries.

Additional sectors which should be mentioned are land transport and transport via pipelines (*H49*) and manufacture of chemically and chemical products (*C20*). The first shows a wide IQR, with the Q3 being on par with energy supply. It lays above average regarding all inflation transmission promoting characteristics in this sample (Figure 3), thereby asserting a predominantly indirect impact on inflation. Land transport (...) arguably achieves systemic importance. The second mention at this point are chemical manufacturing industries. With close to 2.5 forward linkages on average (GO-weighted), it is a highly ubiquitous industry, while also being exposed to notable price volatility. Again, its relevance for final consumption is quite dispersed, with some countries chemical industries pushing the average contribution.

The results identify five sectoral groups as systemically significant, each marked by distinct transmission characteristics. The categorization of clusters closely follows Weber et al. (2024), which reveals significant similarities in the outcome for the US and the EU28. *Housing*, e. g. real estate activities (L68) dominate through consistently high CPI weight, with strong country-level outliers driving inflationary relevance via direct effects. Within the *energy cluster*, petroleum refining (C19) stands out for its volatility and ubiquity as a production input, projecting substantial indirect effects, while electricity and gas supply (D35) derives systemic relevance more evenly across all dimensions, notably

CPI exposure. In the *food cluster*, manufactured food products (C10–C12) contribute through high final consumption relevance, whereas agricultural production (A01) transmits inflation primarily through upstream linkages. *Wholesale and retail trade* sectors (G46–G47) gain relevance through network centrality and consistency, with wholesale marked by broad dispersion and endogenous effects, and retail by direct CPI relevance. *Financial and insurance activities* (K64, K65) show asymmetric systemic relevance, driven by country-specific outliers and primarily indirect effects in financial services, while the opposite holds true for insurance related services. A sixth group, *basic production inputs other than energy* includes chemical manufacturing (C20) and land transport (H49), which do not clearly rank as systemically significant, but exhibit latent systemic significance primarily through high degrees of forward linkages. These findings illustrate that latent systemic price significance arises with each of the three determinants of systemic significance in this model. Thus, sectors identified as such, heavily lean into one of ubiquity, synthetic CPI weight, or price volatility, or reach a combined version of at least two.

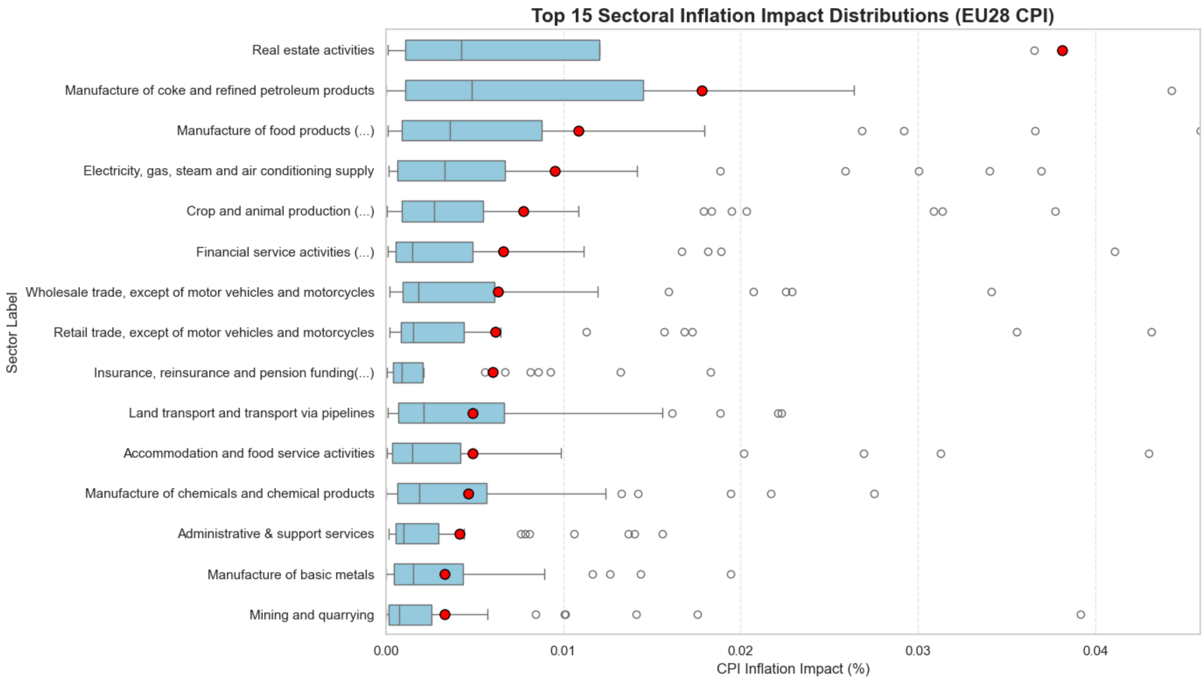


Figure 5: ***Distribution of Sectoral Inflation Impacts on EU28 Countries (2019).*** *The boxplot shows the distribution of total inflation impact (direct plus indirect) for the 15 sectors with the highest average contribution to the synthetic EU28 CPI. Red circles indicate mean values, while boxes represent the interquartile range (25th–75th percentile), the vertical line within the box indicates the median. The x-axis is clipped at the 96th percentile to improve readability by reducing the influence of extreme values. This is why the right whisker of* Real estate activities *is not visible, it reaches outside the clipped area. Note, that the boxplot likely underestimates the interquartile range of EU28 countries as the origin of impact, since the whole sample with 46 countries is plotted. (Own illustration.)*

The distribution of values displayed in the boxplot in figure 5, visually supported by the heatmap in figure 2 reveal differences in the dispersion across countries. While some sectors impact spikes heavily depending on the location, others contribute relevant inflation impact more evenly across several national units. As an example, the nine highest impact real estate activities sectors realize more than 90% of inflation impact on the EU28, assuming all sectors are shocked simultaneously with their given price volatility (Appendix 14). In the same scenario for wholesale trade, 90% are realized by thirteen sectors. For a comparison of the distribution of only EU28 sector impacts, consult appendix 13.

The distribution suggests, that systemic significance might have a structural component: Some sectors are driven by a few, heavy impact national agents, while others have less pointed, but consistent impact. Identifying sectors who assert high influence from only few countries might be especially interesting for targeted policies. Doing so could contain significant inflationary pressure, while only targeting few sectors.

Of course, this interpretation of the distribution and the Pareto charts assumes a collective price shock across all sectors. The likeliness of this scenario differs across industries and regions. Changes in monetary regime by the ECB could influence the price of real estate credit lines, followed by a multinational response in the price system, while a change in regulation does increase housing costs nationally. For more detailed analysis on the nature of price volatility and shocks, see Dawe et al. (2025), Bajraj et al. (2023) and Beck et al. (2009). Being aware of the limitations, the distribution of inflation impact across national sectors does offer interesting insights and considerations for policy design.

While targeting specific sectors can help to maximize effect of targeted policies, knowing geographical inflation trajectories can help improving policy, too. The following section will deal with this.

4.1.2 Geographical Clustering of Inflation Impact

Since the sector perspective was thoroughly investigated, it is now time to focus on the geographical perspective. In their foundational paper, Ipsen et al. (2023) advance a core vs. periphery perspective for the EU , with twelve core EU countries, and sixteen periphery countries, using their respective synthetic CPI shares. Ferreira et al. (2024) build up on the core and periphery logic, with increased granularity. They disaggregate into southern and eastern periphery, and financial hubs.

In contrast to the predefined regional categories used in the previous work, pointing out the core-periphery structure in different granularity, this thesis derives geographical groupings endogenously from the structure of inter-industry price transmission. Specifically, Louvain community detection is applied to the country-level network of absolute model responses to the given sectoral price shocks, calculated using the Leontief price model prior to any CPI weighting. The Louvain algorithm[16], first introduced by Blondel et al. (2008), identifies communities in large networks by optimizing a quantity known as modularity. Modularity measures the difference between the observed density of links within communities and the expected density if links were randomly distributed, thereby quantifying the strength of community structure (for a detailed methodology (a full description of the algorithm and its implementation is provided in Blondel et al. (2008)).

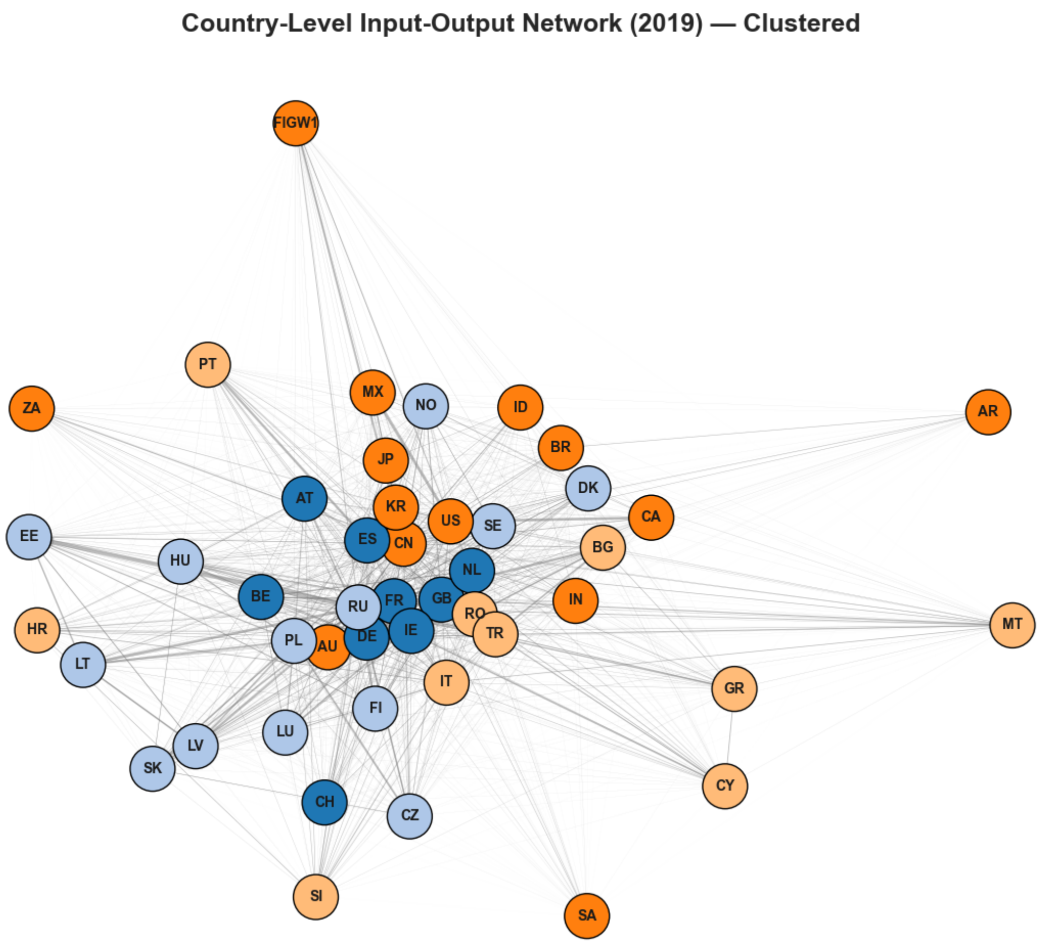


Figure 6: ***Louvain Clustering of Country-Level Inflation Shock Propagation (2019).*** *This network visualization shows how countries are structurally linked through inter-industry price shock transmissions. Each node represents a country, and directed edges capture the total of modeled price impact they have on another. Intra-country effects are excluded. Node colors indicate community membership, and edge thickness reflects the magnitude of shock propagation. The layout is force-directed, emphasizing network structure over geographic proximity. (Own illustration.)*

Each edge in this directed network captures the aggregate price effect that industries in one country exert on those in another. By excluding intra-country flows and focusing on cross-border transmission, the clustering procedure reveals groups of countries whose

[16]The Louvain algorithm is implemented using the `python-louvain` package by Aynaud (2020) (available at https://github.com/taynaud/python-louvain).

economies exhibit a high degree of mutual responsiveness to upstream cost shocks. In this context, a *geographical inflation cluster* is defined as a group of countries where modeled price changes in response to common sectoral shocks are more closely linked within the group than with countries outside it. This approach allows us to identify systemic co-responsiveness in inflation dynamics that emerges from production network topology rather than from consumption structures or policy coordination. While the resulting clusters are expected to reflect elements of the established core-periphery logic, they are not constrained by it and may reveal latent patterns of economic interdependence.

Figure 6 displays the results of the Louvain clustering. The graph is arrange according to the Fruchterman-Reingold-Layout (Fruchterman & Reingold, 1991), which is *force-driven*. A force-driven (or force-directed) layout positions nodes in the network based on a physical analogy: nodes repel each other like charged particles, while edges act like springs that attract connected nodes (Schönfeld & Pfeffer, 2019). This results in a visually intuitive arrangement where more strongly connected nodes are drawn closer together, revealing structural clusters and centrality independent of geographic location.

The graph reveals four clusters in inflation impact:

(i) The *Central and Western European Cluster* (blue, hereafter *Core* or *CO*), including France, Germany, United Kingdom, Spain, Netherlands, Austria, Belgium, Switzerland, and Ireland.

(ii) The *North- and Eastern European Cluster* (light blue, hereafter *North-Eastern Periphery* or *NEP*), with Poland, Russia, Czech Republic, Hungary, Finland, Denmark, Sweden, Norway, Lithuania, Latvia, Estonia, Slovakia[17], and Luxembourg.

(iii) The *Mediterranean and Southern European Cluster* (light orange, hereafter *Southern Periphery* or *SP*), consisting of Italy, Bulgaria, Cyprus, Greece, Croatia, Malta, Portugal, Romania, Slovenia, and Turkey.

(iv) The *Global Cluster* (orange, hereafter *GL*), including the United States, China, and the rest of the world[18].

The prediction regarding the European inflation clusters appears to be largely confirmed by the network results. Overall, the structure supports the core–periphery logic proposed by Ipsen et al. (2023) and Ferreira et al. (2024). The layout reveals a strong centrality of the *Core* cluster: the three largest EU28 economies, Germany, France, and the United Kingdom, are situated in the densely connected center of the graph. These

[17] There is slight ambiguity for Slowakia (SK). The heuristic and random initializations by the Louvain method can produce slightly different partitions. For 10 simulations, Slovakia was assigned to NEP cluster seven times, and to the SEP cluster 3 times. The author assigns it to the NEP cluster.

[18] The reader should note that due to the lack of price data, Saudi Arabia (SA), South Africa (ZA), and Argentina (AR) were not shocked; thus, the edges are only one-way and do not correctly reflect their influence within the price system.

countries show close bilateral links as well as strong connections to the two most influential members of the *North-Eastern Periphery* (NEP): Russia and Poland.

An interesting deviation is Italy, which, despite ranking fourth in both population and GDP within the EU28, is positioned slightly outside the densely packed center and is assigned to the *Southern Periphery* cluster. Spain, by contrast, is part of the Core cluster. This is surprising, since it swaps the countries in the Ipsen et al. (2023) taxonomy, and moves Spain from southern periphery to core compared to the Ferreira et al. (2024) taxonomy. For Italy, this divergence may reflect the longstanding economic divide between northern and southern regions. It is plausible that northern Italy is more closely integrated with the core economies, while the southern regions are structurally closer to the Mediterranean periphery. This could potentially explain cluster identity.

The SP cluster exhibits internal heterogeneity: a few members such as Italy, Romania, and Turkey lie close to the core, while others are located at the network periphery, with a visually high degree of dispersion. This may reflect variation in economic size, openness, or production structure. Countries in the Core cluster likely share characteristics associated with economic centrality—such as large populations, high industrial output, advanced technology, strong financial sectors, or robust growth. While this is not formally tested here, it offers a plausible explanation for their central positioning in the price transmission network.

The cohesion of the Core cluster is visually evident: all member countries are located near or within the dense center of the network. Switzerland is the only member positioned slightly further from the center. The spatial concentration of Core countries reflects a coherent pattern of mutual price transmission. By contrast, SP countries are widely dispersed across the layout, suggesting more fragmented responses to upstream price shocks.

The NEP cluster appears to have two spatially distinct groupings. One, located at the lower left of the graph, includes Russia, Poland, and other Eastern European Countries, and the small Baltic states distinctively at the outskirts. The other consists of the Nordic countries—Sweden, Norway, and Denmark—while Finland is situated in the prior grouping. Thus, while building one cluster, the geographical and structural differences between the Eastern European and the Scandinavian region is visible in the graph.

Overall, the graph supports the existence of geographically and economically coherent inflation clusters. Again, the layout has no geographical component, making the visible, although partial, reproduction of geographic regions in terms of closeness quite interesting. The visualization enables the identification of smaller regional or structural units that exhibit tightly coupled price transmission dynamics. The spatial layout provides an interpretable representation of the systemic price interdependencies in the European production system. Regarding the robustness of the clusters, the clustering was applied over all available years. The observed clusters are generally stable, with a few noteworthy

		Destination			
		NEP	**SEP**	**Core**	**Global**
Origin	**NEP**	9.5082%	0.6011%	0.4961%	0.1448%
	SEP	0.2958%	8.5933%	0.3009%	0.0855%
	Core	0.7719%	0.7707%	6.2808%	0.2764%
	Global	0.5958%	0.4787%	0.5632%	4.3217%
	Total	**11.1717%**	**10.4438%**	**7.6410%**	**4.8284%**

Table 1: ***CPI-Weighted Inflation Impact Between Regional Clusters in 2019.*** *Values indicate the total inflation impact in the destination region's consumer price index caused by shocks originating in the source region.*

changes.

To briefly investigate the regional dependencies, cluster-specific CPI weights have been calculated and applied on the shock results to assess the different inflation impact structures. Specifically, this allows to check spillover effects, and compare the results with Ipsen et al. (2023) and Ferreira et al. (2024). Table 1 shows the inflation spillover from each cluster into another for 2019. Again, simultaneous shocks are assumed and accumulated, similar to the papers mentioned. Focusing on the European regions, the South Eastern Periphery cluster shows the weakest spillover potential. The Core cluster thereby exerts the most pronounced and consistent impact on the two other European regions. The North Eastern Periphery locates between Core and SEP, but has double the impact on SEP than vice versa.

Regarding overall inflation, the two periphery regions do show a higher inflation exposure compared to the Core region. Here, it should be mentioned, that there are large countries included in both, which are not part of EU28, namely Turkey for SEP and Russia for NEP, which does not allow for a direct comparison with the results of Ipsen et al. (2023) and Ferreira et al. (2024). Nevertheless, the results yielded here do support their notion. The European core appears to be less prone towards sectoral inflation pressure compared to the periphery regions. In this price shock scenario, NEP (11.17%) and SEP (10.44%) display significantly higher inflation impact, compared to the core (7.65%). Furthermore, the latter exerts a higher inflationary pressure on the periphery regions, than vice versa, with around 0.77% towards both. The NEP cluster does have a more pronounced impact on the core ($\approx 0.5\%$), compared to the SEP cluster ($\approx 0.3\%$).

4.1.3 Temporal Variation in Systemic Significance

The last aspect of systematically significant prices the author wants to put in perspective is time. It was already highlighted, that systemic significance is not a static concept. The Plot 7 shows the development of weighted inflation impact over the sample time horizon, excluding housing, which has notoriously high and stable impact due to its importance ins household consumption. The ten systemically significant sectors reveal quite volatile developments in their significance. An illustrative example is the British manufacturing of coke and refined petroleum sector. While in 2012 it ranked at the top of this sample regarding inflation impact, it rapidly lost significance over the next ten years. A look at the same sectors of other countries reveals similar patterns for most of them. Apparently, coke and refined petroleum lost a portion of their relevance in the last decade, at least considering household consumption. Now, while this holds true in comparison to the early years of this sample, the significance started to increase again from 2016-2018, followed by a pronounced dip in 2020, when the COVID-19 pandemic hit the global economy. Again, the recent years see a sharp increase.

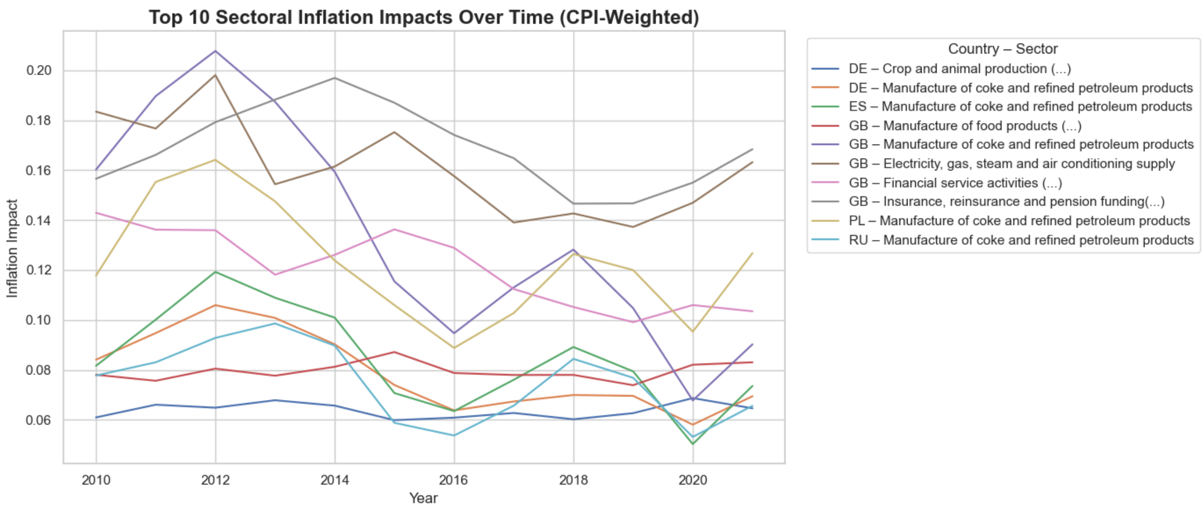


Figure 7: ***Top 10 Sector-Country Inflation Impacts Over Time (2010–2021).*** *The figure shows annual inflation impact trajectories for the most influential sectors, real estate excluded, across countries, based on Leontief price model simulations and EU28 household expenditure weights. (Own illustration).*

The explanation for the 2020 dip is two sided, with demand and supply shocks alike driving down crude oil prices (Bourghelle et al., 2021). Thus, risk of inflation transmission of the industries processing crude oil were practically eradicated, there were no costs to be passed through to industries downstream, while there was also drops in demand. In comparison, the inflation transmission by Financial and insurance related services in GB was practically unaffected; it even rose. Service sectors which were not affected heavily by the restrictions on physical contact were generally less prone to the shocks, some even gained from the situation. Thus, their positions in the price system remained rather stable.

The recovery of the pandemic shock, and then ultimately the Russian invasion in the Ukraine, followed by sanctions on Russian products and commodities, and the natural gas embargo by Russia then created a new dynamic in the industry network. Suddenly, due to the forced substitution away from Russian fossil energy carriers sparked a renewed rise of energy related sectors and their systemic importance in the economy, hence, the price system ("EU to fully end its dependency on Russian energy", 2025). This becomes visible in Figure 7, with a almost uniform increase in inflation impacts across the energy related sectors in this group. Focusing on Energy supply as well as manufacturing of coke and petroleum only, the plots in Appendix 15 and 16 do support this view. Both show a recent increases in their effect on price volatility, in other words, their systematic significance

Overall, observing the temporary development over the full sample shows, that real economic developments are an important driver of systemic significance in the inter-industry network. The data suggests, that also the overall importance of sectoral shocks can change over time. While this is intuitive, it underlines the need of constant evaluation and re-evaluation of (latent) systemically significant sectors, in order to take recent changes in price systems and overall economic dynamics into account.

This concludes the analysis of latent systemically significant prices. Several dimensions of systemic significance have been discussed, to gain an comprehensive view onto the sectoral component in the price system. With focus on EU28, specific systemically significant sectors where identified. Further, it was delved into a more structural view on the industry network, with the goal of identifying structurally important sectors and their characteristics, apart from national specification. The geographical component followed, through identifying regional inflation clusters, and a comparison of transmission tendencies between the regions, paired with the findings of prior research. Lastly, an attempt to display the dynamics of the inter-industry network and resulting volatilities in the price system was made, where some trends were identified. The next section aims to specify a sectoral component of the recent inflation period in more detail: The gas price shock following the war by Russia against the Ukraine.

4.2 Analysis of the Gas-Price Shock in the EU-28

In the first analysis part, we examined the structural role of systemically significant prices in the EU-28 using volatility-based shocks. The following second part will now zoom in on a recent event, which, among several motivational aspects, gave reason for this analysis: the 2021 surge in gas prices following the Russian invasion of Ukraine. This event amplified the renewed urgency of inflation analysis, induced by the COVID-19 pandemic, and serves as a central case study for this thesis. By modeling the associated price shock, its propagation through the EU28 production network can be systematically examined, along with its inflationary effects on consumer prices.

4.2.1 Scenario Design and Modeling Assumptions

The analysis imposes a 500% price increase on the Natural Gas Extraction sector, following the approach of Cucignatto et al. (2023), who base this magnitude on market observations. Specifically, they use Title Transfer Facility (TTF)-based gas-price indices, comparing year-end prices of 2018 (December 31st) with the annual average for 2022 (Cucignatto et al., 2023, p. 266). The TTF, a virtual trading hub based in the Netherlands, serves as a widely used benchmark for European gas prices. While the 500% increase constitutes a stylized assumption, it represents a plausible upper-bound scenario.

Unlike in the foundational paper, the shock is not applied to the 2018 inter-industry network, but to the 2021 structure. This adjustment is motivated by the fact that the industrial configuration in 2021 was the one directly exposed to the observed price shock and is therefore considered the most appropriate reference point for the simulation.

By applying the shock at the extraction level—the most upstream point in the supply chain—it becomes possible to trace its downstream propagation within the inter-industry network. As in the prior analysis, a modified Leontief price model is employed and weighted by the synthetic CPI to estimate the inflationary effects. Given that natural gas extraction is mostly not directly consumed by households, the focus lies solely on indirect (endogenous) transmission, and no direct CPI impact is calculated.

Instead, the origin of the effect is differentiated: Since here, the imports of natural gas are emphasized, the analysis distinguishes between two scenarios to reflect different assumptions about EU gas sourcing:

- *Extra-EU28 scenario*: Only imports from outside the EU28 are shocked.
- *Intra-EU28 scenario*: Both extra-EU and intra-EU natural gas trade flows are shocked.

This distinction is intended to clarify the extent to which EU28 countries depend on natural gas imports from outside the Union. By isolating extra-EU import channels, the analysis provides a more nuanced understanding of the import structure and its significance in the propagation of inflationary shocks. The evaluation of the yielded results focus on three dimensions:

1. **Country-level exposure**: Which Member States experience the highest inflation effects based on their own synthetic CPI?
2. **EU28-wide contribution**: Through which national production systems does the shock most strongly affect the EU28 synthetic CPI?
3. **Sectoral transmission**: Which industries are most critical in mediating the shock's inflationary effects?

This structure facilitates a targeted assessment of transmission mechanisms and informs subsequent policy implications.

4.2.2 Individual Country Impacts and EU28 Inflation Transmission

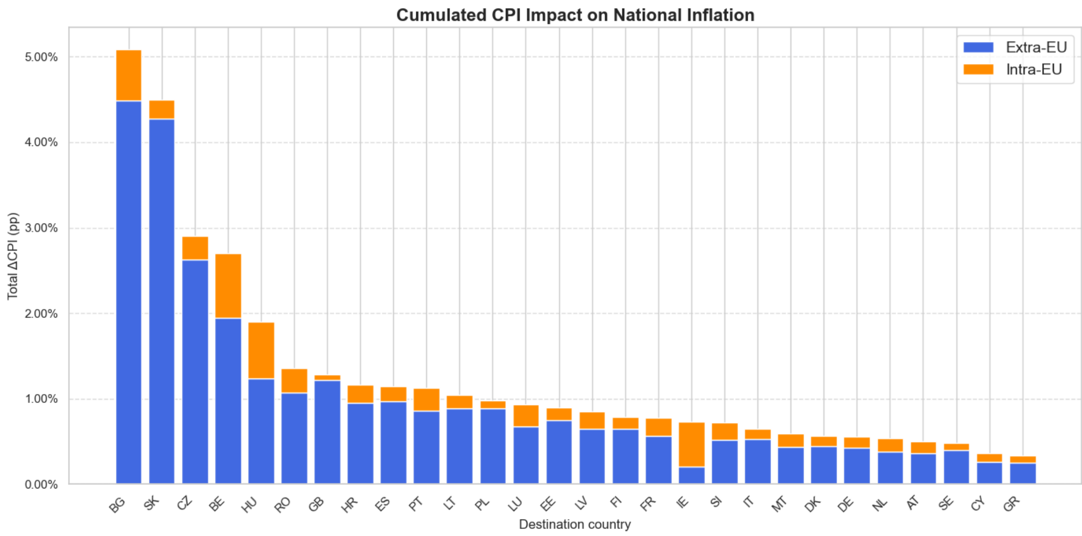


Figure 8: ***Cumulative CPI Impact by Destination Country (Gas Price Shock).*** *This bar chart ranks EU28 countries by inflation exposure to a 500% price shock in Natural Gas Extraction (applied to imports only). Results are based on the 2021 Leontief price model and synthetic CPI weights. (Own illustration).*

Figure 8 ranks Member States by the cumulative CPI effect of a 500 % *import* shock to the Natural Gas Extraction sector. Over all EU-28 countries only a modest share of the consumer-price effect stems from intra-EU gas trade; the bulk is imported from outside the Union. International Energy Agency (2025) statistics confirm, that after 2020, more than four-fifths of the region's gross available natural gas originated in non-EU production, with only three sizeable EU28 extractors–the United Kingdom, the Netherlands, and Romania–supplying meaningful volumes domestically . This supply structure presumably explain the pattern in Fig. 8: The most exposed destinations are Bulgaria ($\approx 5\%$) and Slovakia ($\approx 4.5\%$), followed by the Czech Republic, Belgium, and Hungary. All five are net importers that rely almost entirely on extra-EU gas and possess limited own extraction capacity; consequently the 500 % import shock passes through to household prices largely unmitigated. The individual impacts ranking does mix up, when taking the domestic gas extraction industries into account, as in Appendix 18). Doing this exhibits the pronounced role of reasonably large domestic gas extraction sectors in the price system. Romania and the Netherlands, from a price system perspective, shows the highest dependency on its own gas extraction industry. Romania for instance is already a large gas producer, and aims to become the largest n the EU[19], which supports the high domestic impact resulting from the analysis (International Energy Agency, 2025). Spain and Germany do not have a very relevant domestic gas extraction, therefore, its impact on price levels is negligible. Other countries with relevant price impacts of domestic gas are Great Britain, Poland and Denmark.

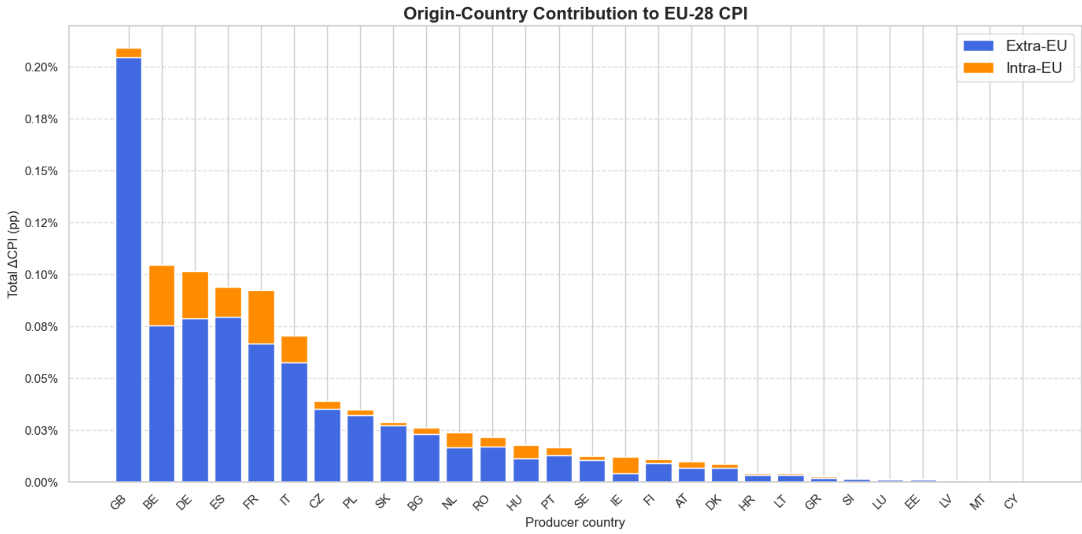


Figure 9: ***Cumulative Sectoral Inflation Contribution to EU28 CPI.*** *The figure shows CPI-weighted sector-level inflation impacts from a 500% natural gas shock. Upstream energy sectors dominate, followed by transmission through intermediate manufacturing and consumption-relevant sectors. (Own illustration).*

In contrast to the high individual impact countries, the large economies, Germany, Great Britain France, Italy and Spain, display comparatively low inflation impacts on the country level, ranging around or below 1%. Their industrial networks, however, mediate a sizable share of the Union-wide burden, shown in Fig. 9. Great Britain is the single most important conduit, transmitting more than 0.2% to EU28 inflation. Belgium, Germany, Spain and France each contribute roughly 0.08–0.1 pp through their industrial networks. These hub economies amplify the shock: they import large volumes of natural gas, then pass price pressures onward via energy-intensive manufacturing and transformation sectors. Taken together, the producer links of GB, DE, BE, IT, and FR transmit roughly half of the total EU28-wide CPI effect. Hence these hub economies work primarily as conduits, not as end-point recipients of inflationary pressure. This is consistent with the regional analysis in Chapter 4.1.2, which shows that large economic hubs occupy central positions in the production network and transmit shocks through extensive forward linkages in the price system.

The findings of this analysis can be contextualized by comparing them to the results presented in Cucignatto et al. (2023), who examine the effects of a 500% gas price shock for

[19]See Gotev (2025)

Italy, France, and Spain. Their baseline scenario, absent price controls of their advanced scenarios, focuses on imported natural gas as the source of the shock. While both studies broadly agree on the transmission channels, especially the role of gas-intensive upstream sectors, several key differences emerge in terms of magnitude and sectoral prominence.

In particular, whereas Cucignatto et al. (2023) report that the electricity and gas supply sector (D35) exhibits the highest price increases (FR: +40%, IT: +37 %, ES: +22.5%), the results in this thesis attribute a comparatively greater role to the petroleum refining sector, which registers the strongest unweighted price impacts in all three countries (FR: +17.7%, IT: +7.9%, ES: +13.8%). Meanwhile, the electricity and gas supply sector in our simulation shows substantially lower increases (FR: +4.3%, IT: +5.9%, ES: +11.7%), reversing the relative ranking between these energy transformation sectors. Furthermore, the price effects in this analysis are considerably more muted than in Cucignatto et al. (2023), pointing to a lower degree of shock transmission—both at the sectoral and aggregate level.

With regard to the countries' individual CPI-weighted and aggregated inflation effects, Cucignatto et al. (2023, p. 267) report increases of 2.41% for France, 3.53% for Italy, and 1.74% for Spain. The results obtained in this thesis are substantially lower, with corresponding values of 0.77%, 0.65%, and 1.14%. Notably, this reverses the relative positioning of Spain: whereas Cucignatto et al. identify Spain as the least affected country, the findings presented here suggest a comparatively higher vulnerability. This deviates more from the absolute level of observed inflation, but it does reflect the relative order of price increases more accurately. As noted by the authors, "the Harmonized Index of Consumer Prices (HICP) grew cumulatively by 11.9% in France, 12.3% in Italy and 13.5% in Spain during the period under scrutiny (2018–2022)" (Cucignatto et al., 2023, p. 267).

Overall, both studies identify upstream energy-intensive sectors as the principal conduits through which gas price shocks propagate through the economy. However, the extent and relative importance of these channels differ—likely reflecting variations in IO databases (ICIO vs. FIGARO), levels of aggregation, and reference years (2018 vs. 2021). These factors may explain the less pronounced inflationary effects observed in this simulation. Nonetheless, the broader conclusion remains consistent: sectors that transform or heavily rely on natural gas inputs represent the most potent levers of inflationary transmission following a major import price shock.

4.2.3 Sectoral Impact and Transmission Channels

While the supply of electricity and gas sector (D35) is surpassed by the manufacture of coke and refined petroleum products (C19) in Italy, France, and Spain individually, the synthetic CPI-weighted results aggregated over all EU28 countries identify D35 as the largest contributor to inflation in EU28. This is in line with expectations, as D35 directly consumes large volumes of natural gas for both transformation and distribution functions. Generally, the CPI-weighted results highlight a clear concentration of inflationary pressure in energy-intensive upstream sectors.

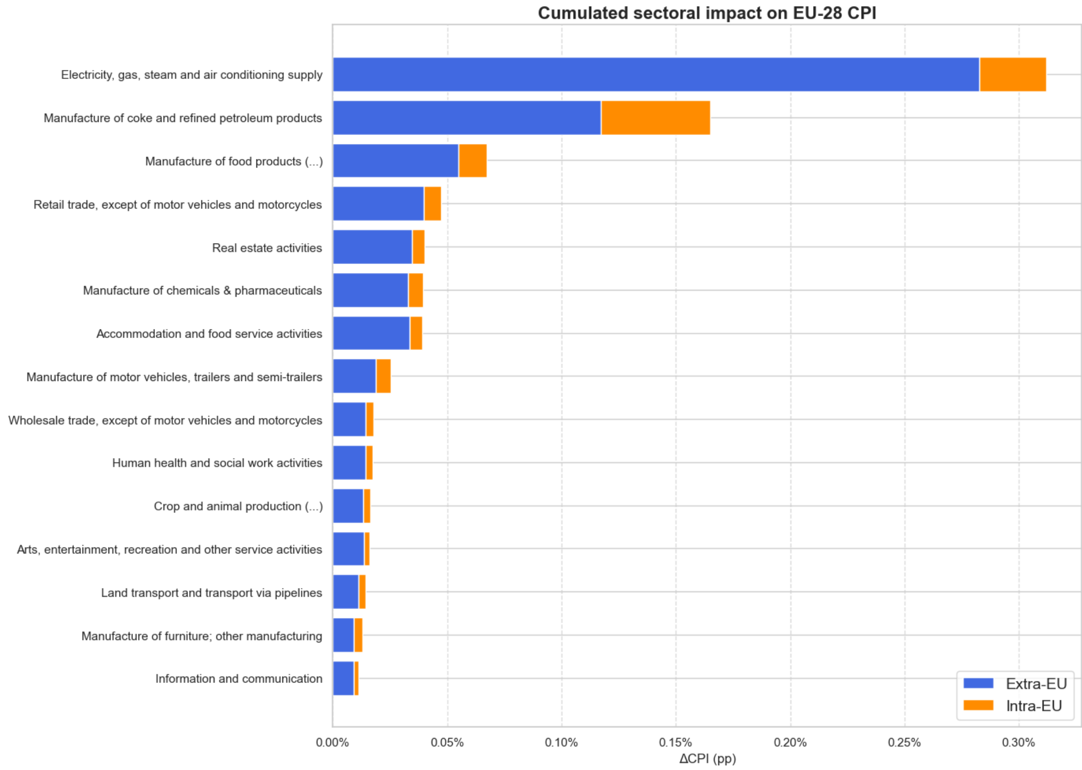


Figure 10: ***Cumulated sectoral impact on EU-28 CPI.*** *Top 15 sectors under a 500 % shock on Natural Gas Extraction. (Own illustration.)*

Closely following is the petroleum refining sector, which also exhibits a substantial CPI-weighted effect. The elevated sensitivity of C19 to a gas price shock reflects the high importance of natural gas in its processes: Gas is the primary fuel source in refining, but also a key input in various transformation processes within this industry (EU MERCI, 2017)[20]. Together, D35 and C19 form the core of price transmission pathways from imported gas shocks to final consumer prices.

Several downstream sectors also display notable, albeit significantly smaller, impacts. Food, beverage, and tobacco manufacturing (C10-C12), retail trade (G47,), real estate activities (L68), manufacture of chemicals and pharmaceuticals (C20_C21), and accommodation and food service activities (I) register lower but non-trivial CPI contributions, in relation to other sectors. These impacts likely arise primarily through indirect channels: cost pass-through from energy-intensive suppliers or higher energy-related operational expenditures.

Again, the relevance of sectoral weights in household consumption baskets must be

[20]The EU-MERCI project features informative technical reports, fact-sheets and more information for a range of the most important industrial products, with a focus on energy usage and efficiency. See http://www.eumerci.eu/the-project/ for in-depth information on production processes and energy usage.

carefully considered. For instance, the notable contribution of the real estate activities sector (L68) does not primarily reflect direct exposure to energy costs in production. Its relevance is based on the relevance of housing in private household consumption, as pointed out in chapter 4.1.1. The average unweighted effect in relative prices ranks 44th of 50 sectors in this aggregation, with a price change of $\overline{0.36\%}$. Vice versa, applying the synthetic CPI weights does cover up important price transmitters, with low relevance for household consumption directly. Many manufacturing industries need natural gas as an important primary input. These industries act as intermediaries of the gas price shock; price increases in their outputs pose increased costs for downstream sectors, which define large parts of consumer price inflation. The relevance of manufacturing industries in the price system becomes visible, when looking at the average, unweighted impacts in table 2.

Table 2: ***Average unweighted price change by sector across EU28 (Top 10)***

Row Labels	Average of Price Change
Electricity, gas, steam and air conditioning supply	11.46%
Manufacture of coke and refined petroleum products	8.87%
Manufacture of basic metals	6.15%
Manufacture of chemicals and pharmaceutical products	3.53%
Manufacture of other non-metallic mineral products	3.47%
Manufacture of fabricated metal products	2.28%
Manufacture of furniture and other manufacturing	1.92%
Manufacture of paper and paper products	1.85%
Mining and quarrying, except natural gas	1.83%
Manufacture of rubber and plastic products	1.80%

The overall structure of results is consistent with a cost-push inflation mechanism. Upstream sectors–most notably in manufacturing–play a key intermediary role in transmitting energy cost shocks throughout the economy. While these sectors are only marginally represented in final consumption baskets, their exposure to natural gas inputs leads to pronounced price increases, which are then passed on to downstream sectors more directly linked to household demand. This cascading effect explains why some consumer-facing industries register inflationary pressure without themselves being major users of natural gas. In this sense, the observed inflationary pattern reflects an indirect propagation dynamic shaped by the position of sectors within the production network.

Taken together, the findings support three stylized facts outlined in the preceding chapters: (i) import dependence shapes vulnerability—Member States with domestic gas extraction capacities tend to exhibit greater resilience, particularly when price increases in the extraction sector are retained or dampened through internal pricing mechanisms; (ii) network centrality amplifies transmission—industrially integrated economies redistribute external shocks across borders, even when their own direct CPI exposure remains limited;

and (iii) regional asymmetries persist—Eastern and South-Eastern European economies continue to show heightened structural sensitivity to imported energy shocks. These empirical regularities will guide the subsequent policy discussion in the concluding chapter.

Chapter 5

Discussion and Conclusion

5.1 Discussion of Key Findings

The core finding of this thesis is that sectoral price shocks have an asymmetric impact on inflation, with a small subset of sectors disproportionately influencing aggregate prices. In particular, systemically significant sectors can transmit outsized inflationary effects when shocked. Sectors qualifying as systemically significant are those highly connected upstream in production networks or those with large weights in consumer expenditure. For example, energy-related industries (upstream providers of critical inputs) and housing or food sectors (with large CPI shares) emerged as pivotal nodes. Price increases in these sectors cascade through input-output linkages or directly raise consumer prices, resulting in a broad-based impact on the cost of living. This result aligns with recent research showing that micro-level shocks in key industries can amplify into significant macroeconomic volatility (Acemoglu et al., 2016; Ferreira et al., 2024; Ipsen et al., 2023; Weber et al., 2024). It underscores that inflation is not purely a uniform macro phenomenon, as suggested by many orthodox economists, but often the outcome of sector-specific disturbances propagating unevenly through the economy's structure.

From a geographical perspective, this analysis identifies distinct inflation clusters emerging *after* the transmission of price shocks. The community detection analysis of the endogenous shock propagation results, building on the idea of predetermined clusters by Ipsen et al. (2023), also picked up by Ferreira et al. (2024), reveals a cluster taxonomy that diverges from their work, especially with regard to Italy and Spain. While the Louvain method was used for community detection, it is optimized for large-scale networks, and alternative methods maybe better suited for the relatively small EU28 network examined here.

Importantly, the analysis revealed that the systemic importance of industries evolves over time, as illustrated in Figure 7. A comparison with earlier network analyses (e.g. Ipsen et al., 2023) indicates notable shifts in the ranking of critical sectors. For instance, real estate activities have gained even more prominence in driving inflationary risk since 2014, whereas refined petroleum manufacturing has become somewhat less dominant in the latest data. Such changes reflect structural economic transformations (technological shifts, policy changes, globalization patterns, etc.) over the past decade. In line with Weber et al. (2024, p. 301), this implies that policymakers must continuously reevaluate the industrial structure of countries, regions or the EU: today's key inflation-driving sectors may not be tomorrow's. Put differently, systemic significance is dynamic across time and space.

The 2021–2022 gas price shock serves as an illustrative case study of how an upstream price surge transmits through the economy and manifests in inflation. In the model scenario, a spike in natural gas prices propagates via higher input costs for electricity, manufacturing of coke and refined petroleum, and gas-intensive manufacturing industries, like basic metal and chemicals, ultimately feeding into consumer prices for a wide array of goods. The simulated transmission was, again, uneven across regions: Eastern European countries showed particularly strong inflationary exposure to the gas shock. Such regional asymmetries suggest a uniform policy response is suboptimal – a theme revisited in the policy recommendations.

Comparing the simulated inflation outcomes to actual inflation during 2022 reveals considerable discrepancy. The input-output price model, which captures direct and indirect cost pass-through, estimated a certain increase in the overall price level. However, actual headline inflation in the EU far exceeded the pure cost-shock simulation for the gas scenario. This is likely the result of additional disruptions in the price system, as well as the aftermath of the pandemic. Yet, this gap likely reflects profit-driven inflation dynamics, as firms raised mark-ups in response to the shock (Cucignatto et al., 2023; Hansen et al., 2023; Hayes & Jung, 2022; Konczal & Lusiani, 2022; Rabensteiner et al., 2025; Weber & Wasner, 2023). The concept was explained in detail in section 2.3.1. While this thesis doesn't delve deeply into profits, it underscores the importance of industry-level pricing decisions in inflation. If sectoral pricing decisions are not considered relevant for overall price levels, profit squeezing has no important impact on inflation. Papers like Ferreira et al. (2024), Ipsen et al. (2023), and Weber et al. (2024) and this thesis establish sectoral pricing decisions as relevant, enabling profit inflation as a somewhat complementary inflation amplifier.

According to the recent contributions, many firms did not just pass on higher costs to maintain margins; instead, they increased their mark-ups, resulting in price hikes larger than cost increases alone would justify. The gas shock provided a cover and impetus for this behavior: energy companies enjoyed windfall profits at the shock's onset[21], and downstream firms in concentrated markets found room to widen their profit margins ("sellers' inflation"). Thus, the gas price shock case demonstrates both the power and limits of cost-push inflation modeling–it captures the structural propagation well, but a full account of inflation needed to consider changes in pricing power and market behavior under stress, apart from other limitations laid out in section 3.6.

This interpretation is supported by recent empirical studies. For instance, Cucignatto et al. (2023) and Rabensteiner et al. (2025) show that profit expansion played a central

[21]These profits were based on price setting principles in energy markets, like the merit-order principle, rather than being decided deliberately.

role in driving inflation across Europe, particularly in concentrated sectors such as energy and food. These findings align with Post-Keynesian theories of *conflict inflation*, which view inflation as the outcome of distributional struggles between capital and labor. As Hein (2024) argues, inflation arises when competing claims on income—wages and profits—exceed the productive capacity of the economy. This shift in income shares–widening profit margins alongside lagging wages–epitomizes the conflict over distribution emphasized by Post-Keynesian theory. In the recent episode, profit claims clearly dominated: profit claims clearly dominated: real wages declined while profit margins expanded alongside rising prices. Wages, therefore, did not cause inflation—they absorbed it. While real wages have begun to recover, workers have not yet regained the purchasing power lost during the inflationary episode (Janssen & Lübker, 2024). Recognizing this distributional conflict is key to understanding the full mechanics of recent price dynamics and points toward a differentiated set of policy responses.

Taken together, these insights reveal inflation as a layered process: cost-push pressures may trigger price increases, but the trajectory is shaped by the pricing behavior of firms and the underlying distributional dynamics. This has major implications for policy design, as elaborated in the policy implications section.

5.2 Policy Implications and Recommendations

The finding that much of the inflation was rooted in sector-specific shocks and profit dynamics means that conventional aggregate monetary policy is a blunt and potentially misguided tool in this context. Across 2022-2023, central banks responded to high inflation with interest rate hikes aimed at cooling demand (English et al., 2024; European Central Bank, 2022). However, demand suppression has a very limited effect on cost-push and profit-driven inflation. In the scenarios at hand, shocks occurred at times when economies had spare capacity (e.g. during the COVID downturn or early recovery), so raising interest rates would not address the supply constraints or pricing power issues fueling prices. Instead, contractionary monetary policy in such circumstances risks doing more harm than good. It deepens economic downturns and delays recovery by reducing aggregate demand and discouraging investment. Yet it fails to directly counteract an energy shortage or to limit monopolistic pricing. The social costs are significant: households suffer twice—first from the higher prices and then from job and income losses due to the induced slowdown. Indeed, real wages had already been eroded by the inflation shock, effectively draining household purchasing power. Further tightening then curtails any rebound in wages or employment, cementing a loss in living standards. In short, using broad interest rate tools to fight this type of inflation imposes unnecessary welfare costs and may prove ineffective or even counterproductive in tackling structurally rooted price pressures. A one-size-fits-all anti-inflation approach is ill-suited when the inflation itself is heterogeneous in origin, and/or not characterized by actual features of persistent inflation.

Instead, the analysis promotes targeted, structural policy responses. This may include fiscal and regulatory tools such as price stabilization measures, windfall profit taxes, or interventions in concentrated markets. Such measures could diversify policy toolboxes for inflation containment. They seem to be better suited for the nature of the shocks analyzed here. Going forward, a comprehensive inflation governance framework is needed—one that integrates monetary, fiscal, and structural tools in response to specific inflationary dynamics. Crucially, input–output analysis can support this process by identifying systemically significant sectors and tracing transmission channels. In doing so, it enables more precise and effective interventions, tailored to the origin and structure of each inflation episode.

5.3 Theoretical Reflections and Limitations

This discussion highlights that inflation is a heterogeneous phenomenon best contained with a broad toolkit. The macroeconomic narrative of “too much money chasing too few goods” does not fully explain an episode where specific supply shortages and corporate pricing shaped outcomes. Instead, the evidence aligns with post-Keynesian and structuralist perspectives that stress the role of cost shocks, distributional conflict, and sectoral structure in inflation dynamics. These views posit that inflationary processes are structurally embedded – influenced by industry characteristics, market power, and institutional context – rather than uniform across the economy. This work, by bridging micro-level sector analysis with macro inflation outcomes, reinforce this stance. The integration of micro and macro aspects enriches understanding: an aggregate inflation rate is the end result of myriad sectoral price movements, themselves contingent on network linkages and power relations. This approach moves beyond the aggregative lens and embraces a more granular understanding of price stability. Supporting the sectoral price shock vs. actual and persistent inflation hypothesis made in the early chapter of this thesis, recent price shock episodes seem to be contained by now. Wages did not exert outsized pressure to prompt a transition into a persistent process of wages and prices/profits.

Finally, it is important to acknowledge this thesis’s limitations. The input–output model used is essentially static and lacks a time dimension, so it captures one-off shock impacts but not the dynamic adjustment path. We did not model how a transitory spike could evolve into persistent inflation through second-round effects like wage adjustments or adaptive expectations. In reality, as discussed, a supply shock can set off sequential phases (markup expansion, then wage responses) that sustain inflation beyond the initial impulse. This analysis provides a *snapshot* of transmission channels and impact magnitudes, but cannot tell how quickly prices revert or how long the shock’s effect lasts. Likewise, the IO framework assumes fixed technical coefficients and unlimited capacity at base prices, which might not hold if the shock is large or prolonged. These simplifications mean the results should be interpreted with caution: they indicate exposure and potential impact, but not a full dynamic forecast. Future research could integrate a temporal dimension or hybrid models (e. g. input-output within a macroeconomic simulation) to examine how sectoral shocks translate into inflation over time, including feedback loops (such as wage-price spirals or policy responses). Further, an integration of market power, i. e. centrality, within sectors would be quite interesting. It could provide further information on the role of highly oligo- or monopolistic markets for price transmission.

The central message stands: understanding inflation requires constant assessment of its origins, with respect to micro and macro level indicators. By combining macroeconomic outcomes with microeconomic foundations, policymakers can better distinguish the nature of inflation they are dealing with–and select the appropriate tools accordingly. This nuanced approach is essential for effective stabilization in an era of overlapping supply shocks and complex production networks. Ultimately, the discussion and recommendations here advocate for a more targeted, structure-aware inflation policy regime – one that acknowledges the diversity of inflationary processes and addresses the root causes in each case, rather than relying on blunt aggregates alone.

5.4 Conclusion

This thesis examined the structural transmission of sectoral price shocks to aggregate inflation in the EU28, using an input–output modeling approach informed by recent episodes of inflation volatility. The analysis demonstrated that a small number of systemically significant sectors—particularly those with strong upstream linkages or large CPI shares—exert disproportionate influence on inflation dynamics. Moreover, the systemic importance of sectors evolves over time, highlighting the need for continuous reassessment of economic structures.

The 2021–2022 gas price shock served as a case study, revealing both strong geographic asymmetries and limitations of traditional cost-push models in explaining the full extent of inflation. Evidence from recent literature points to profit-led inflation dynamics, where firms expanded margins under the cover of external shocks–a phenomenon consistent with post-Keynesian conflict inflation theory. These findings challenge the adequacy of aggregate monetary policy responses and point toward a need for more targeted, sector-specific policy tools.

This work contributes to a more granular understanding of inflation, bridging microeconomic production structures and macroeconomic outcomes. While focusing on micro level aspects, i. e. inflation triggered by sectoral price shocks, the analysis itself takes on a macro perspective, looking at sectoral averages of the EU28, country clusters and temporal dynamics. Potential future research by myself or others could build on this by incorporating time dynamics, firm behavior and structures within markets more explicitly, allowing for the study of inflation persistence, feedback loops, and the role of market power. Also, a more pointed analysis of certain regions or sectors might yield more specific insights. In an era of overlapping shocks, a structure-aware inflation governance framework is not just desirable–it is essential.

Appendix

A.1 Use of AI Assistants

Generative AI was used for coding, reviewing text, and using LaTeX more efficiently. The model was not used for direct writing, but as an assistant for critique, grammar, and flow improvement in english language.

A.2 NACE Coding and Aggregation

Table 3: *NACE Codes with descriptions and their aggregation for WIOD and EXIOBASE datasets.*

NACE Rev.2 Code	Figaro Description	WIOD Aggregation	EXIOBASE Aggregation
A01	Crop and animal production, hunting and related service activities		
A02	Forestry and logging		
A03	Fishing and aquaculture		
B	Mining and quarrying		B_gas and B_nongas
C10–C12	Manufacture of food products, beverages and tobacco products		
C13–C15	Manufacture of textiles, wearing apparel, leather and related products		
C16	Manufacture of wood and cork products (excl. furniture); straw articles		
C17	Manufacture of paper and paper products		

NACE Rev.2 Code	Figaro Description	WIOD Aggregation	EXIOBASE Aggregation
C18	Printing and reproduction of recorded media		
C19	Manufacture of coke and refined petroleum products		
C20	Manufacture of chemicals and chemical products		C20_C21
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations		C20_C21
C22	Manufacture of rubber and plastic products		
C23	Manufacture of other non-metallic mineral products		
C24	Manufacture of basic metals		
C25	Manufacture of fabricated metal products, except machinery and equipment		
C26	Manufacture of computer, electronic and optical products		
C27	Manufacture of electrical equipment		
C28	Manufacture of machinery and equipment n.e.c.		
C29	Manufacture of motor vehicles, trailers and semi-trailers		
C30	Manufacture of other transport equipment		
C31_C32	Manufacture of furniture; other manufacturing		
C33	Repair and installation of machinery and equipment		
D35	Electricity, gas, steam and air conditioning supply		
E36	Water collection, treatment and supply		

NACE Rev.2 Code	Figaro Description	WIOD Aggregation	EXIOBASE Aggregation
E37–E39	Sewerage, waste management, remediation activities		
F	Construction		
G45	Wholesale and retail trade and repair of motor vehicles		
G46	Wholesale trade, except of motor vehicles		
G47	Retail trade, except of motor vehicles		
H49	Land transport and transport via pipelines		
H50	Water transport		
H51	Air transport		
H52	Warehousing and support activities for transportation		
H53	Postal and courier activities		
I	Accommodation and food service activities		
J58	Publishing activities		J
J59_J60	Motion picture, video, broadcasting activities		J
J61	Telecommunications		J
J62_J63	IT and computer services		J
K64	Financial service activities		
K65	Insurance and pension funding		
K66	Activities auxiliary to financial services		
L	Real estate activities		
M69_M70	Legal and accounting; management consultancy		M_N
M71	Architectural and engineering activities		M_N
M72	Scientific R&D		M_N
M73	Advertising and market research		M_N

NACE Rev.2 Code	Figaro Description	WIOD Aggregation	EXIOBASE Aggregation
M74_M75	Other scientific and veterinary activities		M_N
N77	Rental and leasing activities	N	M_N
N78	Employment activities	N	M_N
N79	Travel agency and reservation services	N	M_N
N80–N82	Security, landscape, admin	N	M_N
Q86	Human health activities	Q	Q
Q87_Q88	Residential care and social work	Q	Q
R90–R92	Arts, libraries, gambling	R_S	R_S
R93	Sports and recreation	R_S	R_S
S94	Membership organisations	R_S	R_S
S95	Computer and household repair	R_S	R_S
S96	Other personal services	R_S	R_S
T	Household services		
U	Activities of extraterritorial organisations		

A.3 Construction of Regional Consumption Weights

To enable CPI aggregation at the level of regional country groups (e.g., EU28, core, periphery), regional consumption weights are constructed from the sectoral final household consumption data available in the FIGARO input-output tables. The procedure follows a simple additive method:

1. For each country c in the regional group R, the final household consumption vector is extracted from the final demand block, specifically from the column labeled `P3_S14`.

2. These vectors are summed across all countries $c \in R$ to obtain the total regional household consumption by sector:

$$C_i^{\text{region}} = \sum_{c \in R} C_i^c$$

3. Sectoral CPI weights w_i^{region} are then calculated by dividing each sector's consump-

tion by the total regional consumption:

$$w_i^{\text{region}} = \frac{C_i^{\text{region}}}{\sum_j C_j^{\text{region}}}$$

This method ensures internal consistency of weights across different regional samples. The derived weights can be used to estimate the synthetic CPI impact for each region and allow for direct comparison of inflationary effects across sectors and regional groupings.

A.4 Robustness of Nulling Domestic Gas Coefficients

The price experiment is evaluated in *changes*, not levels. With

$$\Delta P_E \;=\; \left(I - A_{EE}^{\top}\right)^{-1} A_{XE}^{\top}\, \Delta P_X, \qquad \Delta P_X = \operatorname{diag}(\delta_1, \ldots, \delta_g),\ \delta_g \in \{0, s\}, \tag{B.1}$$

robustness follows from two elementary facts:

(i) **Zero shock ⇒ zero response.** If $\Delta P_X = 0$, the rightmost factor in (B.1) vanishes, hence $\Delta P_E = 0$.

(ii) **Nulled exogenous coefficient ⇒ no propagation.** Let $A_{XE}^{*} = M \odot A_{XE}$ with a mask M that sets selected elements of A_{XE} to 0 (domestic flows, and, under the *extra-EU* scenario, generally intra-EU flows), leaving A_{EE} untouched. Substituting A_{XE}^{*} into (B.1) gives

$$\Delta P_E^{*} \;=\; \left(I - A_{EE}^{\top}\right)^{-1}\left(A_{XE}^{\top} \odot M^{\top}\right)\Delta P_X.$$

Each column with $m_{rs} = 0$ becomes the zero vector, *irrespective of* δ_g; the nulled coefficient contributes nothing.

Therefore a term can affect ΔP_E only if it is (a) non-zero in A_{XE} *and* (b) multiplied by $\delta_g \neq 0$. Setting domestic (and, for the *extra-EU* run, intra-EU) gas coefficients to zero leaves the price-change solution (B.1) unaltered.

A.5 Additional Visual Results Supporting Systematically Significant Prices Identification

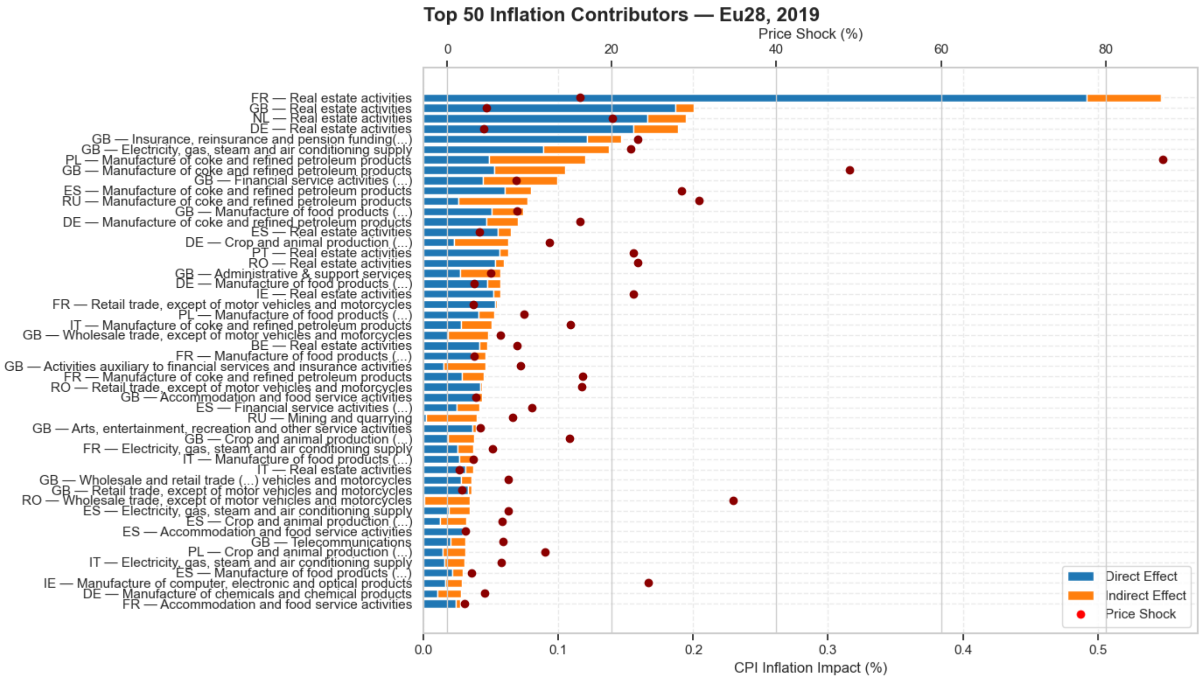


Figure 11: ***Top 50 Sectors by Average Inflation Impact on EU28 CPI.*** *Sectors ranked by average CPI-weighted impact across all EU28 countries in 2019. (Own illustration.)*

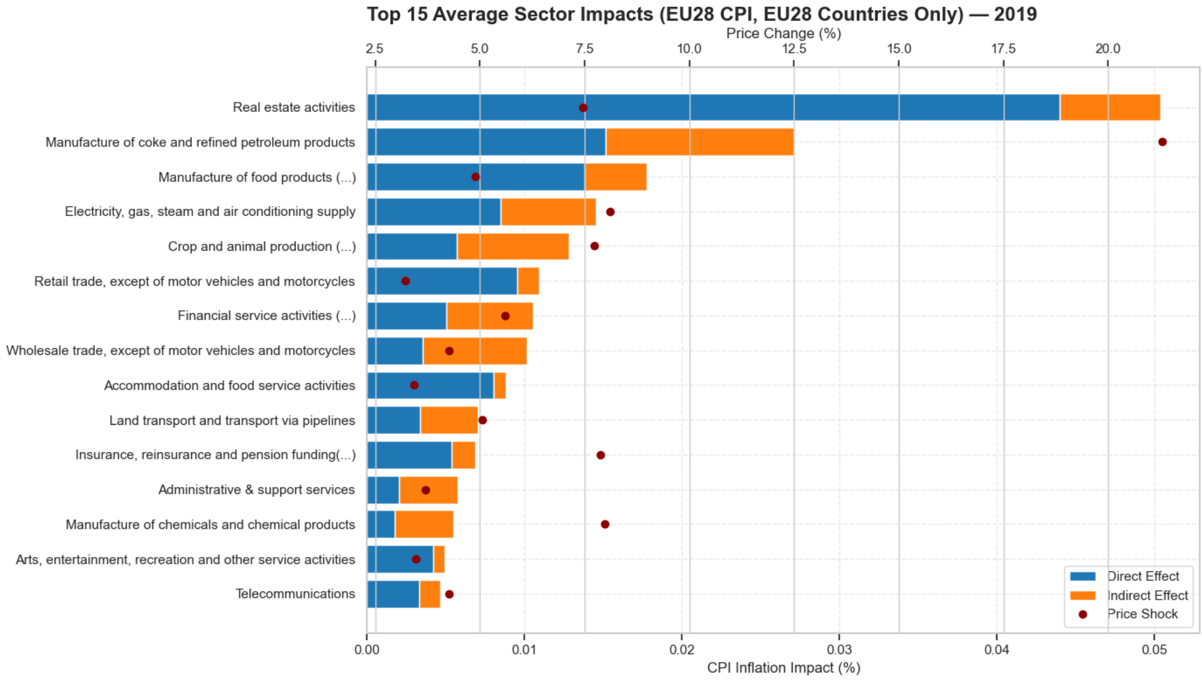


Figure 12: ***Top 15 Sectors by Average Inflation Impact on EU28 CPI originating from EU28.*** *Sectors ranked by average CPI-weighted impact across all EU28 countries in 2019. (Own illustration.)*

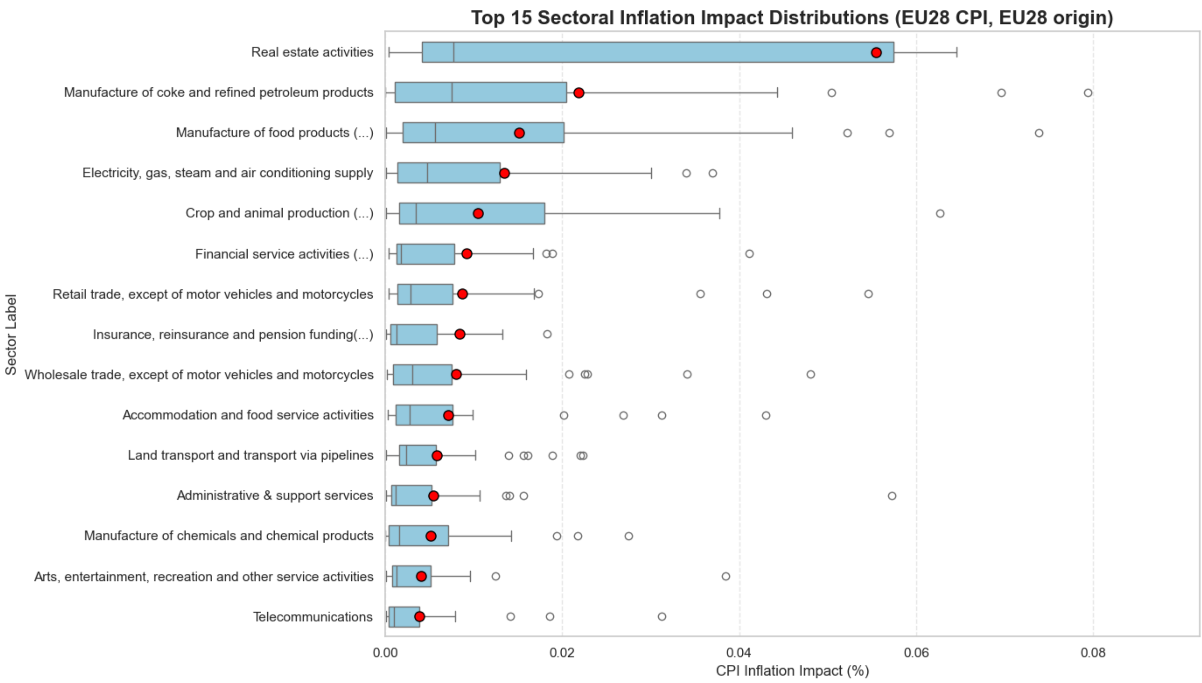


Figure 13: ***Distribution of Sectoral Inflation Impacts in the EU28.*** *Boxplots show the spread of total impacts across countries for the top sectors in 2019. (Own illustration.)*

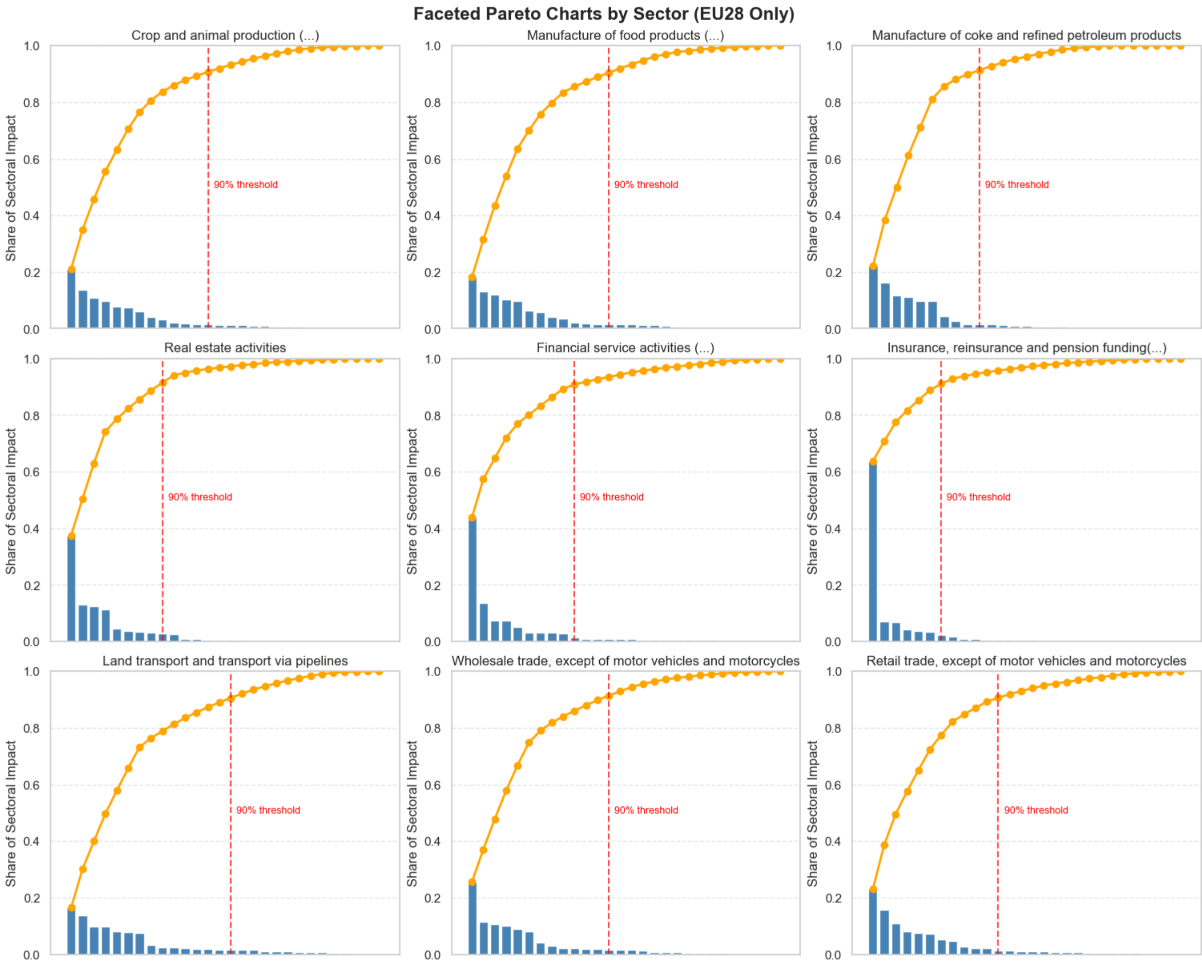


Figure 14: ***Volatility vs. Inflation Contribution of EU28 Sectors.*** *Faceted plot highlighting sectors with high inflation relevance and/or price volatility. (Own illustration.)*

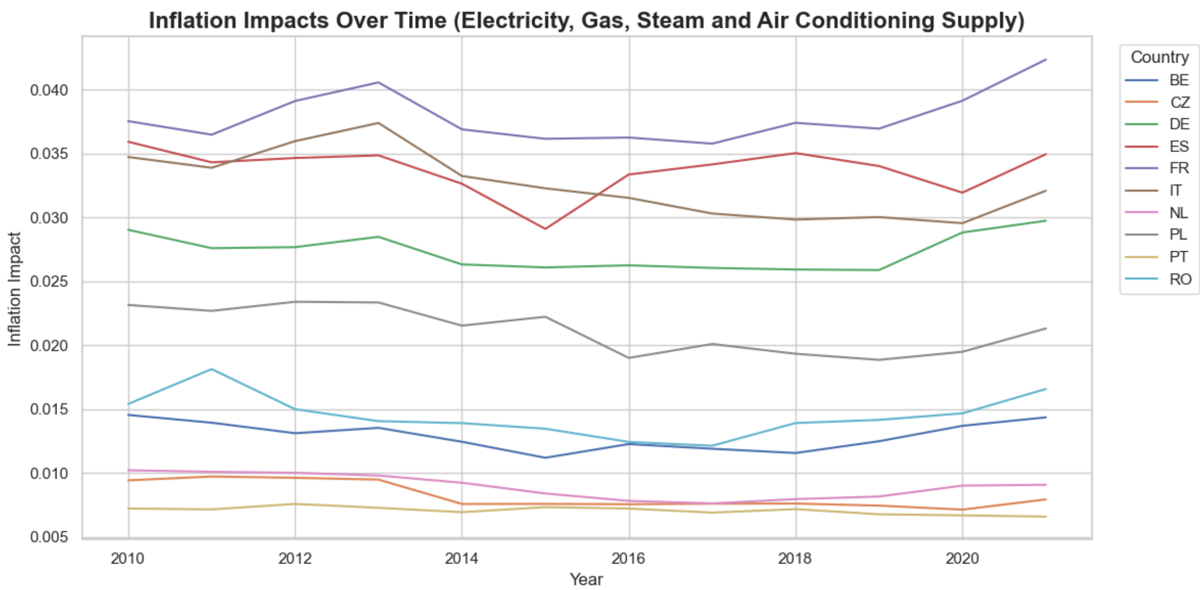


Figure 15: ***Inflation Impact of Electricity and Gas Supply Over Time.*** *CPI-weighted inflation impact of sector D35 from 2010 to 2021. (Own illustration.)*

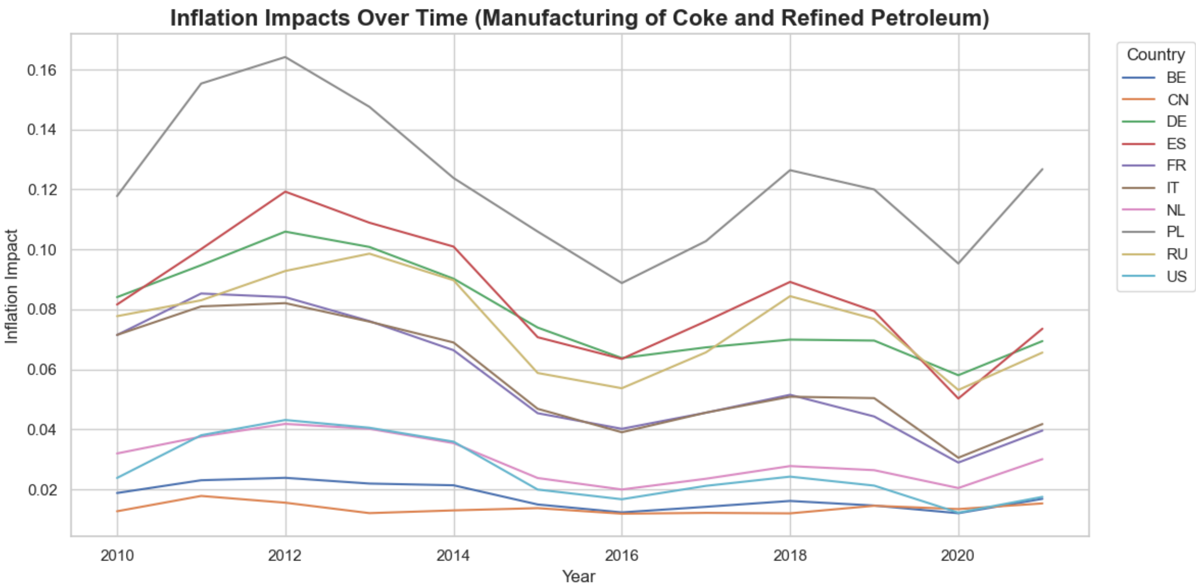


Figure 16: ***Inflation Impact of Refined Petroleum Manufacturing Over Time.*** *Time series of sector C19's CPI-weighted inflation impact in the EU28. (Own illustration.)*

A.6 Additional Visual Results Supporting Gas Price Shock Analysis

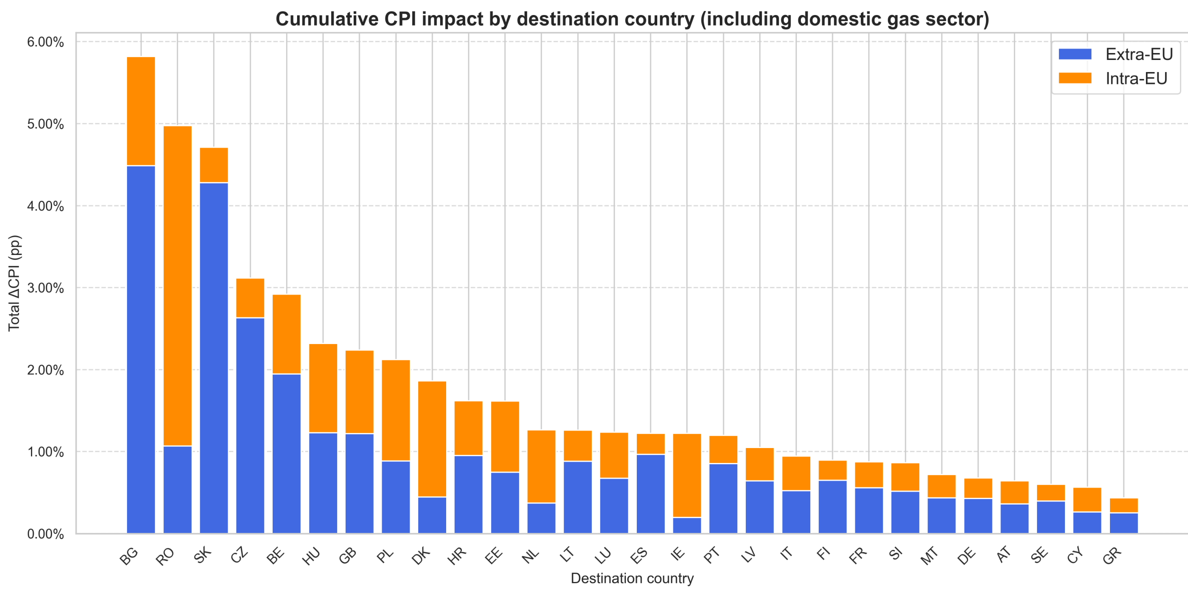
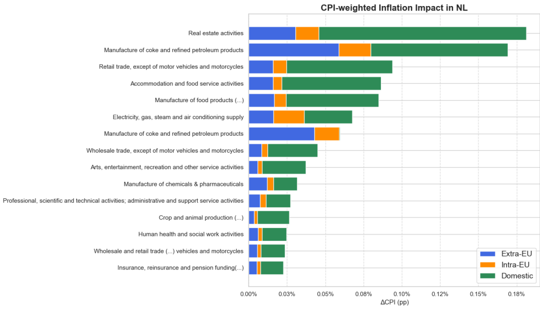
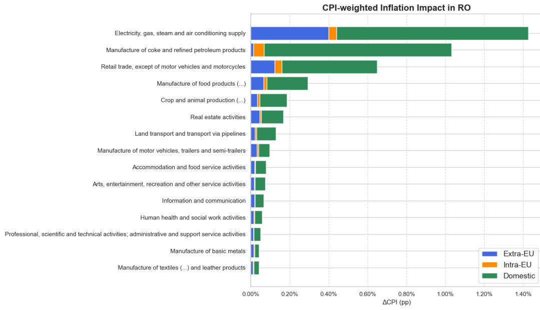


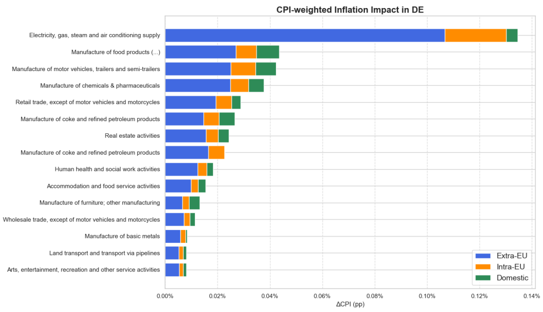
Figure 17: ***CPI-Weighted Shock Transmission Including Domestic Destinations.*** *Inflation impact from a 500% gas shock across EU28 countries, including domestic extraction. (Own illustration.)*



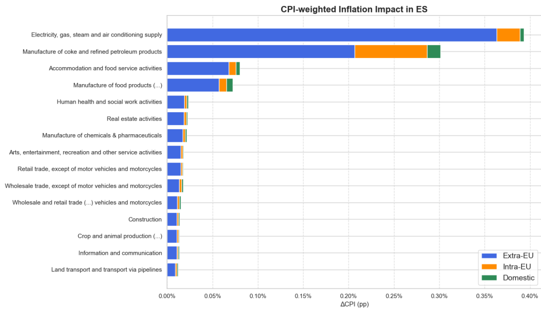
(a) *Netherlands*



(b) *Romania*



(c) *Germany*



(d) *Spain*

Figure 18: ***Country Comparison of Impacts Including Domestic Gas Production.*** *The two at the top, Netherlands and Romania, have sizable domestic production, opposite to Germany and Spain. (Own illustration)*

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